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(54) **GATE STRUCTURES IN SEMICONDUCTOR DEVICES**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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claimer.

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Related U.S. Application Data

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(51) **Int. Cl.**
H01L 21/3115 (2006.01)
H10D 30/01 (2025.01)
(Continued)

(52) **U.S. Cl.**
CPC **H01L 21/3115** (2013.01); **H10D 30/024**
(2025.01); **H10D 30/6211** (2025.01);
(Continued)

(58) **Field of Classification Search**

CPC H01L 29/513; H01L 21/3115; H01L
21/823462; H01L 29/401; H01L 29/7851
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,105,490 B2 8/2015 Wang et al.
9,236,267 B2 1/2016 De et al.
(Continued)

FOREIGN PATENT DOCUMENTS

DE 102019129773 A1 5/2021
KR 20150034221 A 4/2015
(Continued)

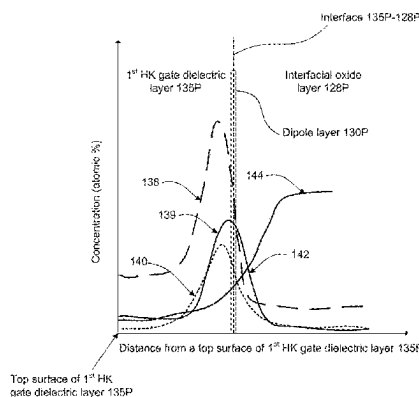
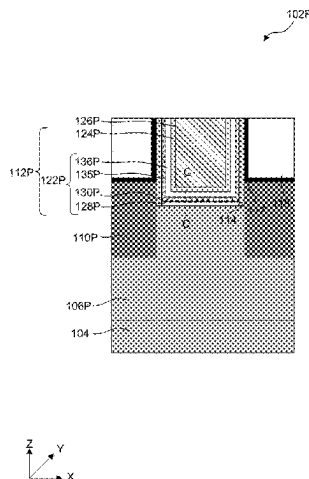
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(57) **ABSTRACT**

A semiconductor device with different configurations of gate structures and a method of fabricating the same are disclosed. The method includes forming a fin structure on a substrate, forming a gate opening on the fin structure, forming an interfacial oxide layer on the fin structure, forming a first dielectric layer over the interfacial oxide layer, forming a dipole layer between the interfacial oxide layer and the first dielectric layer, forming a second dielectric layer on the first dielectric layer, forming a work function metal (WFM) layer on the second dielectric layer, and forming a gate metal fill layer on the WFM layer. The dipole layer includes ions of first and second metals that are different from each other. The first and second metals have electronegativity values greater than an electronegativity value of a metal or a semiconductor of the first dielectric layer.

20 Claims, 23 Drawing Sheets



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H10D 64/68 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
- (52) **U.S. Cl.**
 CPC *H10D 64/017* (2025.01); *H10D 64/681*
 (2025.01); *H10D 64/691* (2025.01); *H10D*
84/0144 (2025.01); *H10D 84/0193* (2025.01);
H10D 84/038 (2025.01)
- (56) **References Cited**
 U.S. PATENT DOCUMENTS
- | | | | | | | | |
|-----------------|---------|--------------|-------------------|---------|-------------|-------|----------------|
| 9,236,300 B2 | 1/2016 | Liaw | 2007/0128736 A1 * | 6/2007 | Chang | | H01L 21/31645 |
| 9,406,804 B2 | 8/2016 | Huang et al. | 2010/0099245 A1 * | 4/2010 | Hyun | | 257/E21.208 |
| 9,443,769 B2 | 9/2016 | Wang et al. | | | | | H01L 21/823842 |
| 9,520,482 B1 | 12/2016 | Chang et al. | | | | | 257/E21.409 |
| 9,548,366 B1 | 1/2017 | Ho et al. | 2012/0049297 A1 * | 3/2012 | Takeoka | | H01L 29/517 |
| 9,576,814 B2 | 2/2017 | Wu et al. | | | | | 257/410 |
| 9,653,548 B2 | 5/2017 | Dewey et al. | 2017/0005175 A1 * | 1/2017 | Song | | H01L 27/085 |
| 9,831,183 B2 | 11/2017 | Lin et al. | 2019/0088498 A1 * | 3/2019 | Wang | | C23C 14/48 |
| 9,859,386 B2 | 1/2018 | Ho et al. | 2020/0119019 A1 * | 4/2020 | Tsai | | H01L 21/823871 |
| 10,784,260 B2 | 9/2020 | Park et al. | 2020/0135475 A1 * | 4/2020 | Cheng | | H01L 21/823842 |
| 10,847,637 B2 | 11/2020 | Chiu et al. | 2021/0028285 A1 * | 1/2021 | Chang | | H01L 29/401 |
| 11,374,090 B2 | 6/2022 | Chen et al. | 2021/0082706 A1 * | 3/2021 | Yu | | H01L 21/823807 |
| 11,581,416 B1 * | 2/2023 | Chang | 2021/0118874 A1 | 4/2021 | Huang | | |
| | | | 2021/0134974 A1 * | 5/2021 | More | | H01L 21/28088 |
| | | | 2021/0217870 A1 * | 7/2021 | Lee | | H01L 29/66545 |
| | | | 2021/0242092 A1 | 8/2021 | Chen et al. | | |
| | | | 2021/0257258 A1 * | 8/2021 | Huang | | H01L 27/0886 |
| | | | 2021/0375629 A1 * | 12/2021 | Lai | | H01L 21/823462 |
| | | | 2022/0223695 A1 * | 7/2022 | More | | H01L 27/092 |
| | | | 2022/0344354 A1 * | 10/2022 | Lin | | H01L 21/823842 |
| | | | 2023/0057278 A1 * | 2/2023 | Chang | | H01L 29/66545 |
| | | | 2023/0058221 A1 * | 2/2023 | Chang | | H01L 27/0924 |
- FOREIGN PATENT DOCUMENTS
- | | | |
|----|---------------|--------|
| KR | 20190013403 A | 2/2019 |
| KR | 20190076251 A | 7/2019 |
| KR | 20210053163 A | 5/2021 |
- * cited by examiner

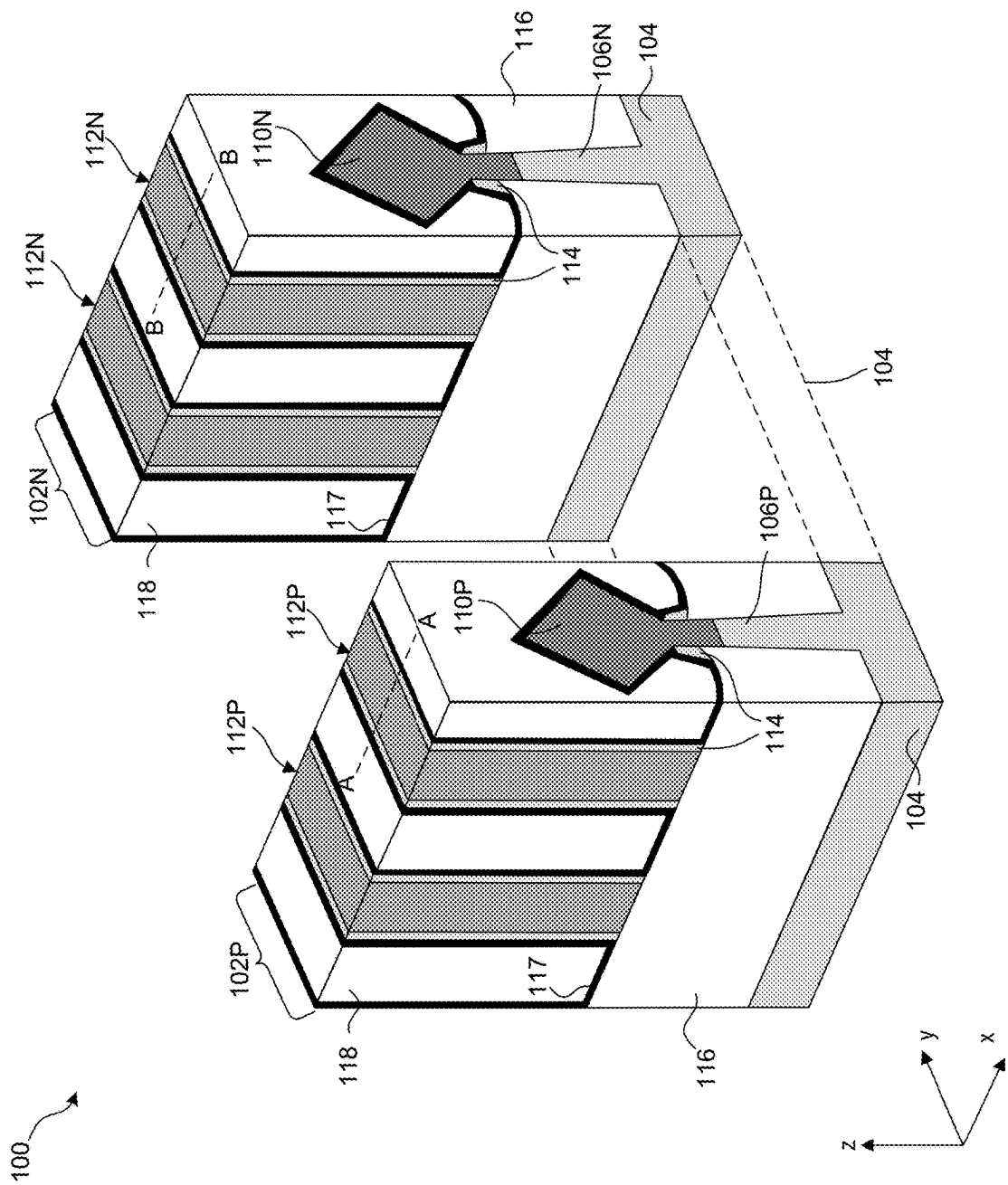


Fig. 1A

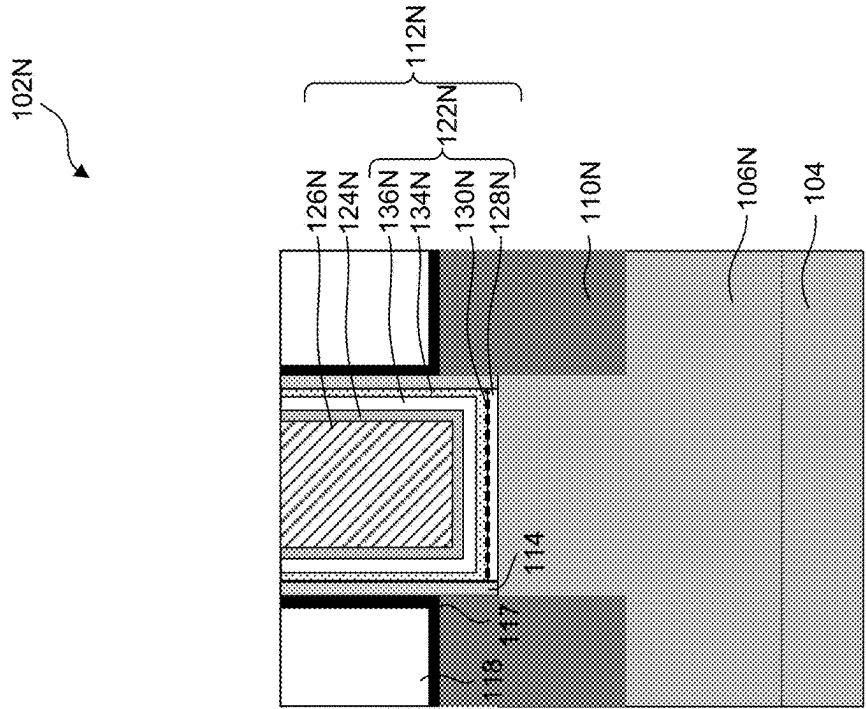


Fig. 1C

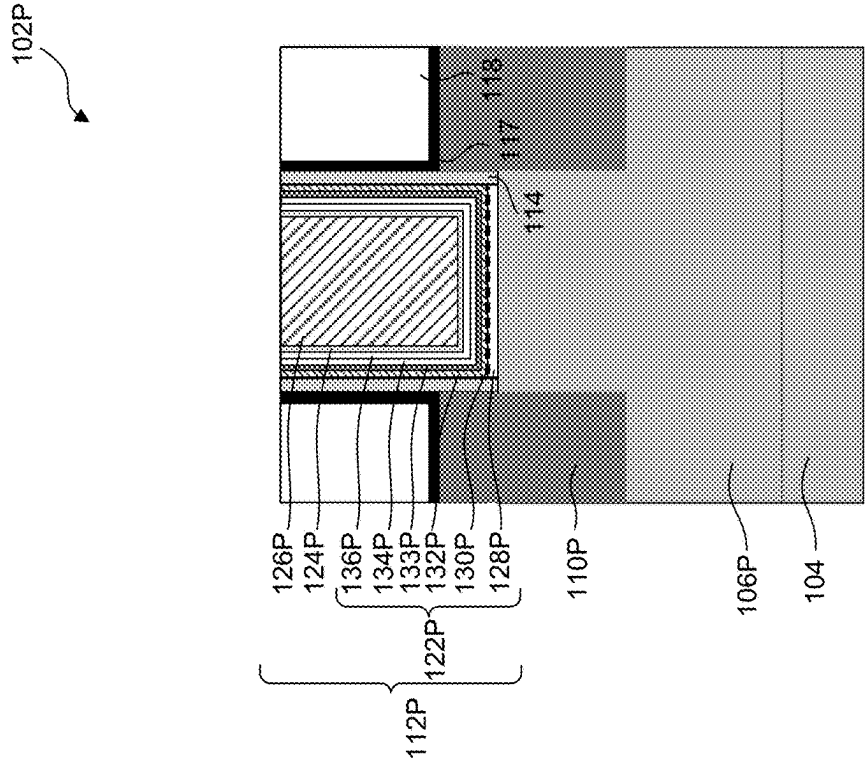


Fig. 1B

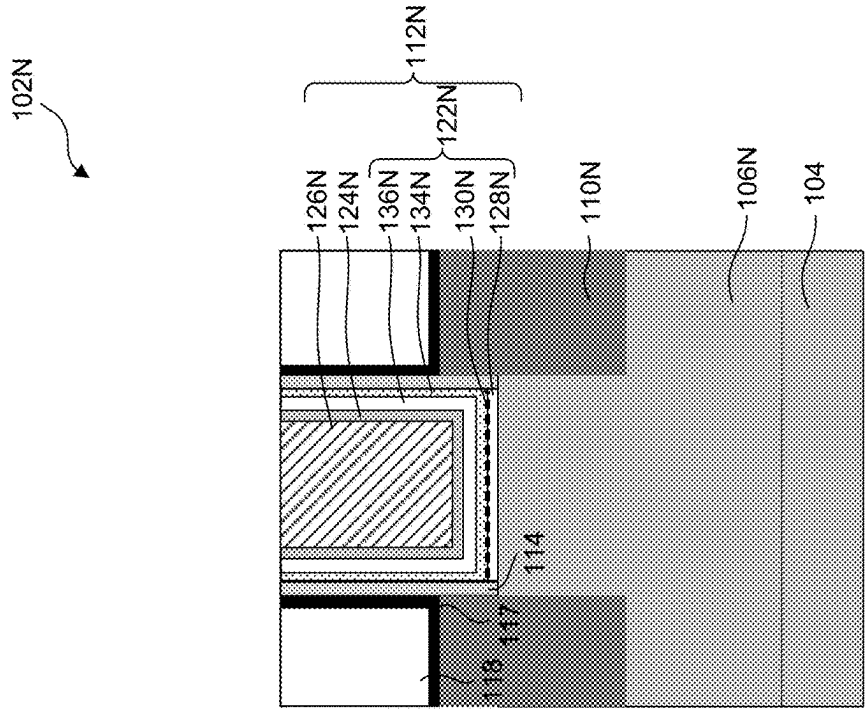


Fig. 1E

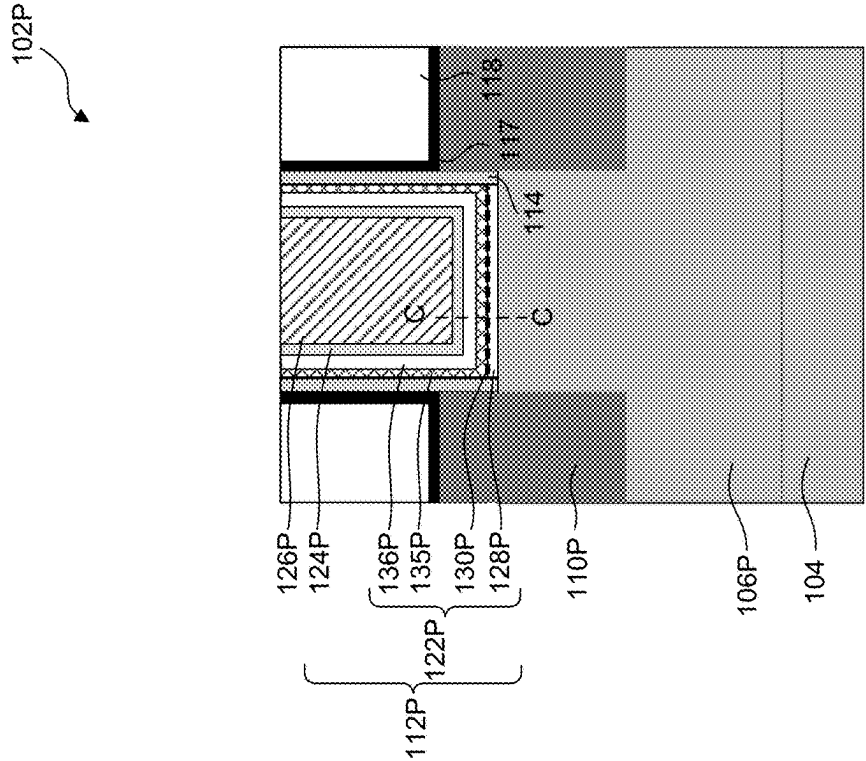
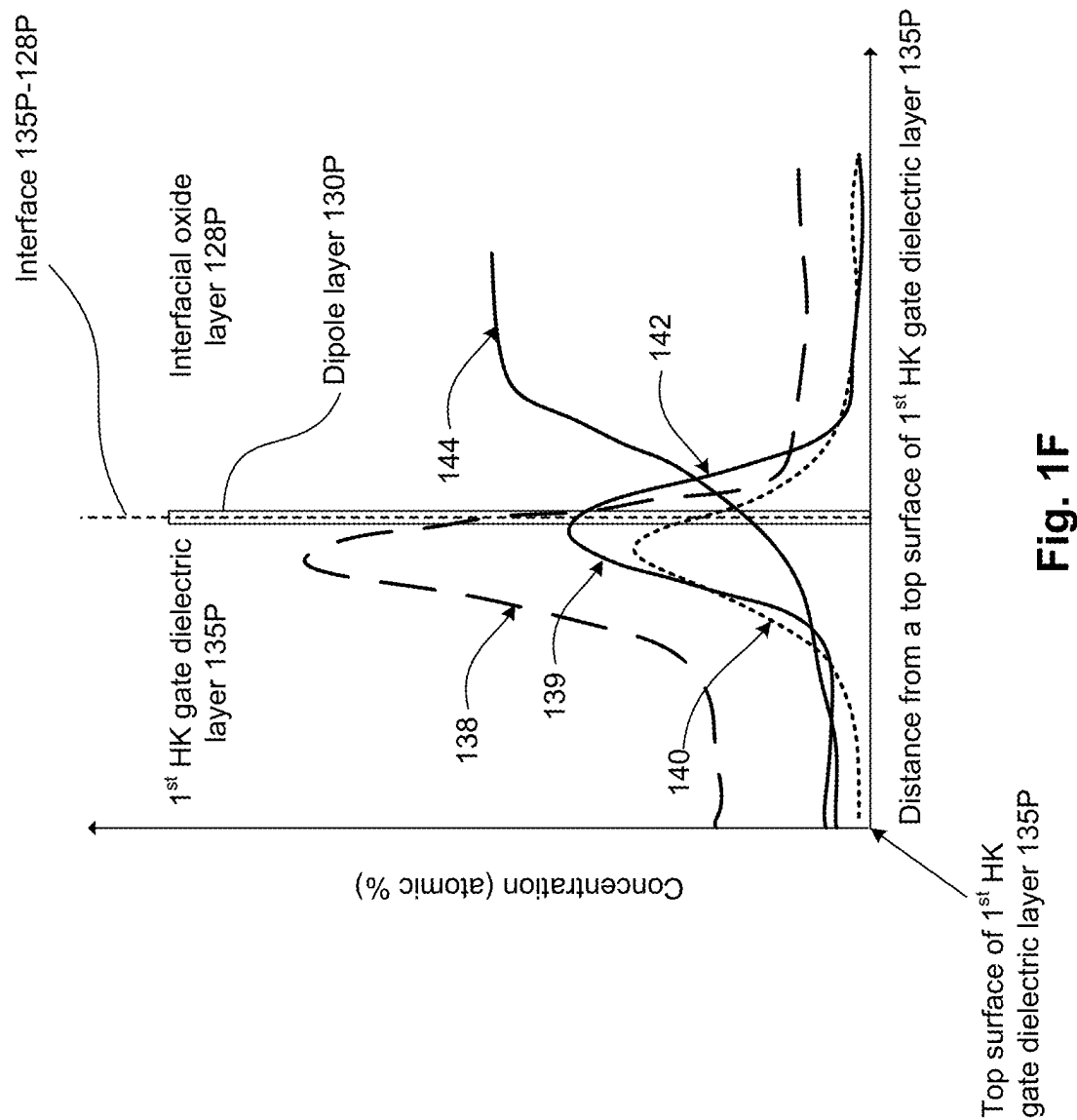


Fig. 1D



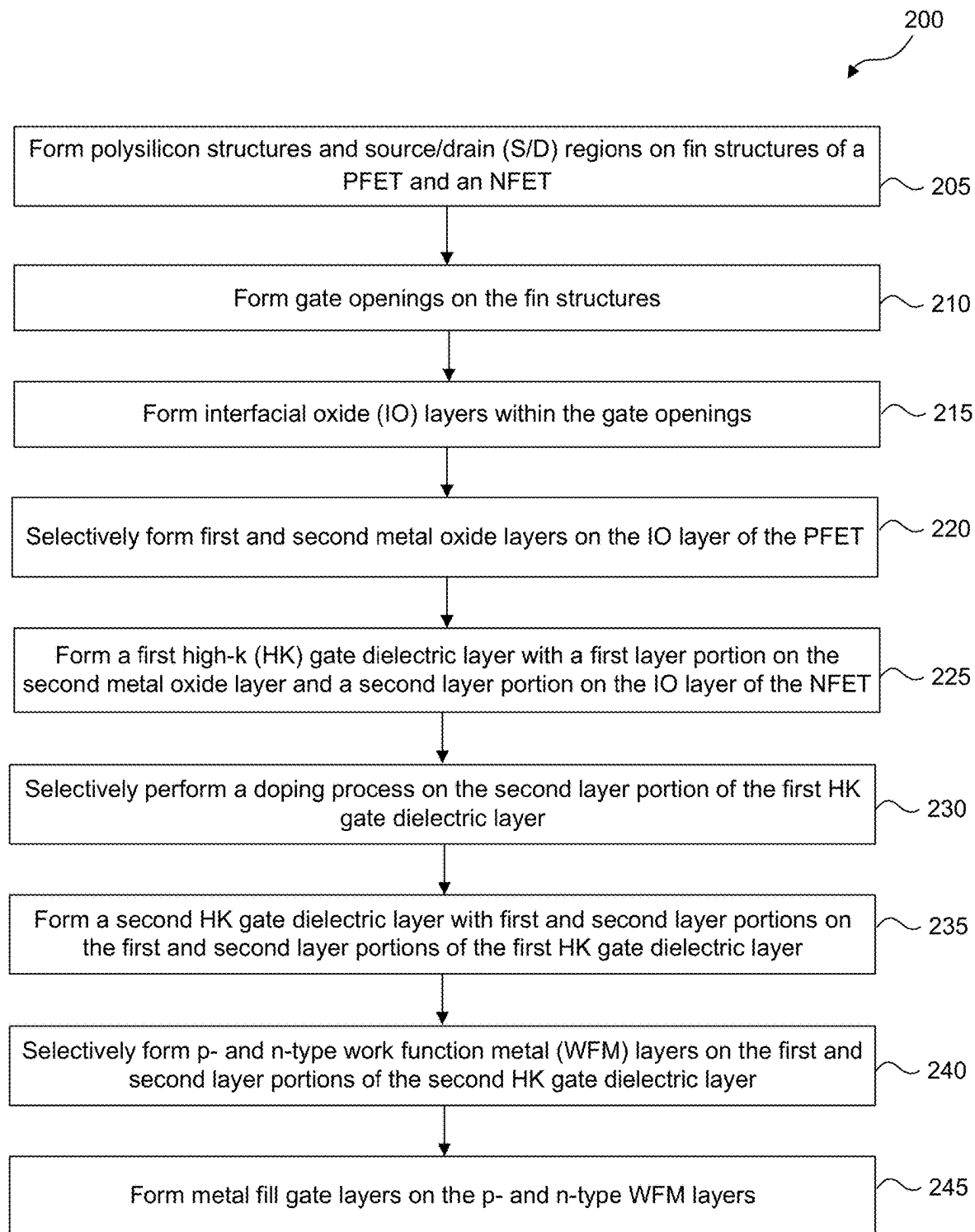


Fig. 2

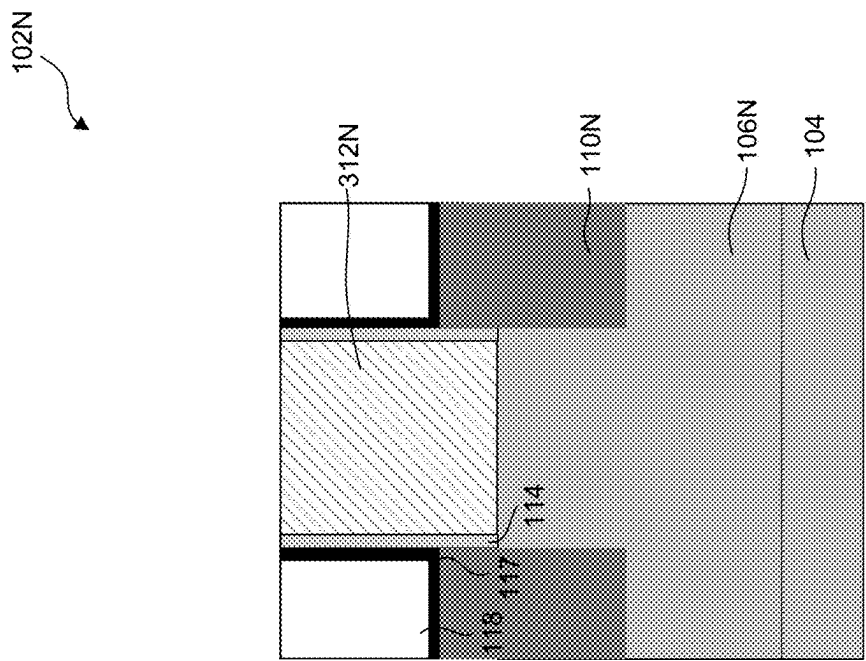


Fig. 3B

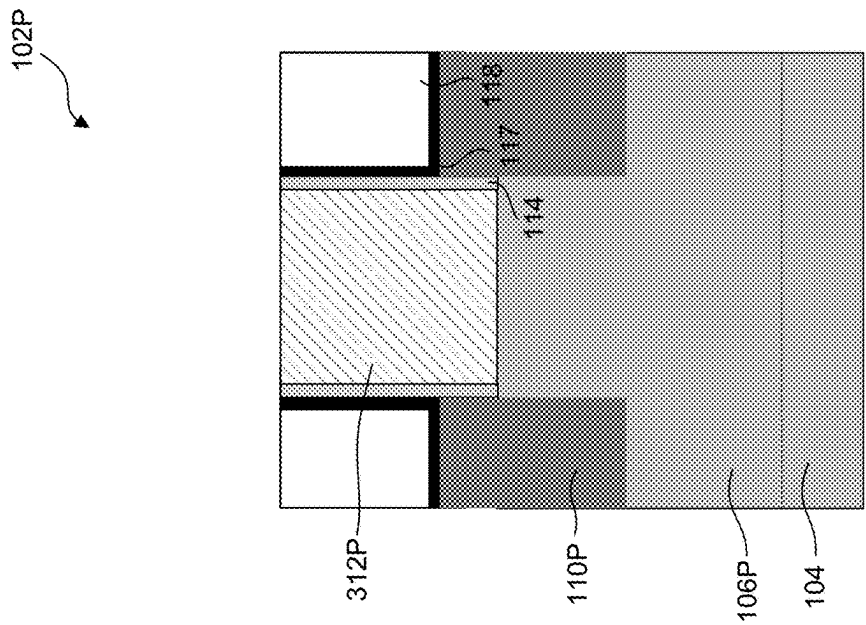


Fig. 3A

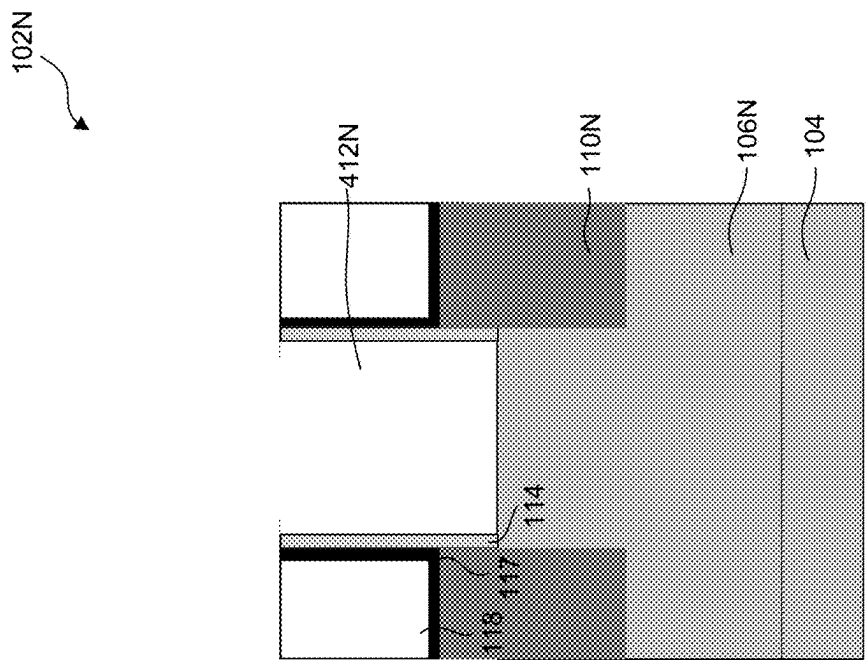


Fig. 4B

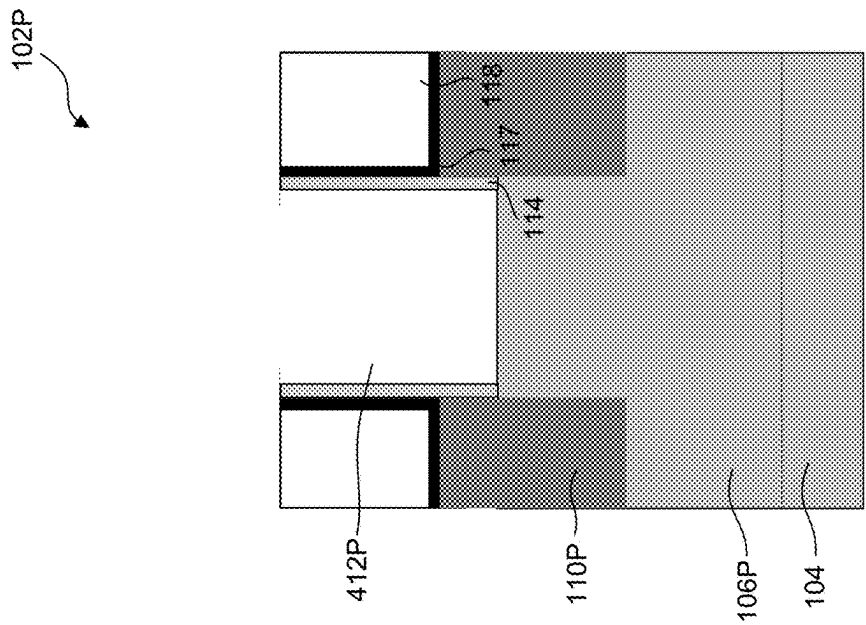


Fig. 4A

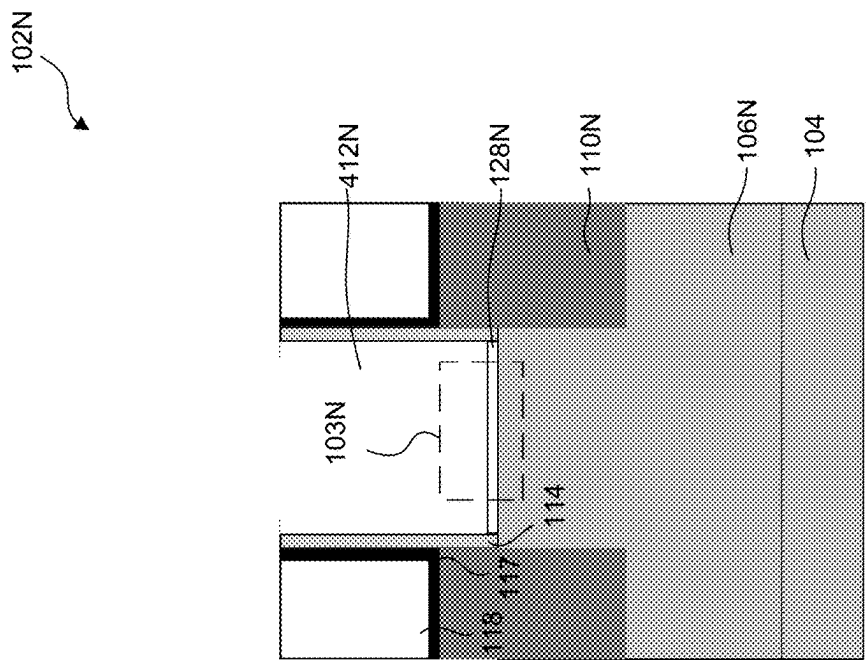


Fig. 5B

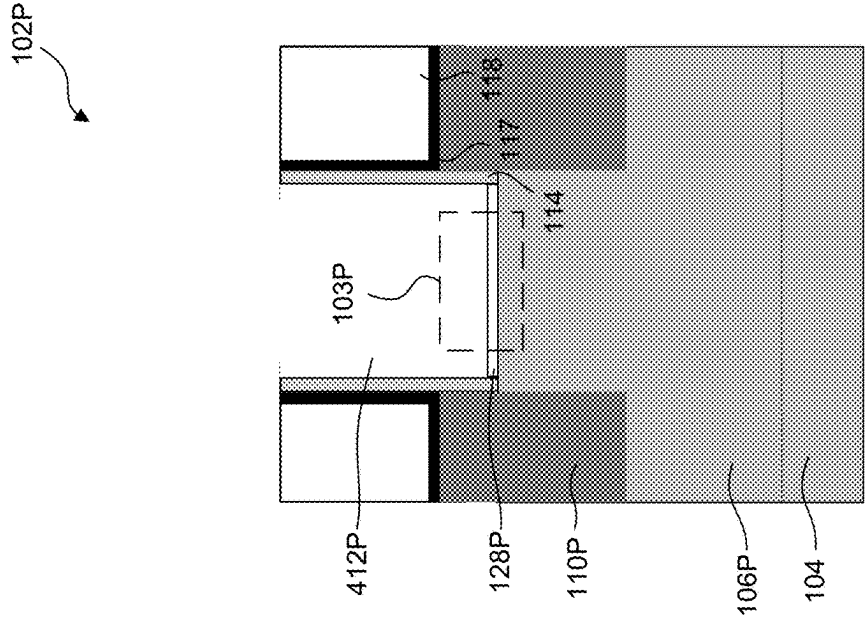


Fig. 5A

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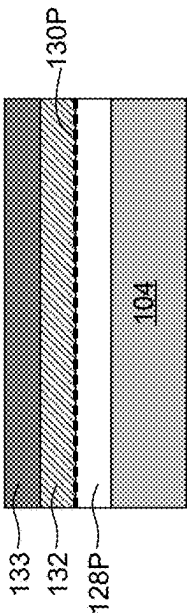
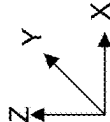


Fig. 6A



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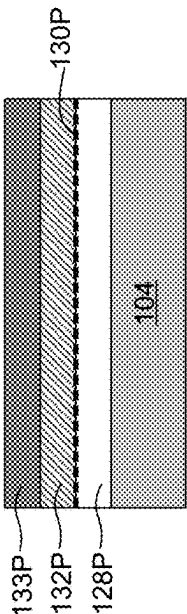
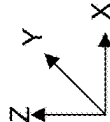


Fig. 7A



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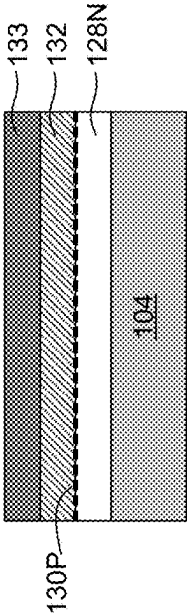
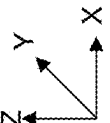


Fig. 6B



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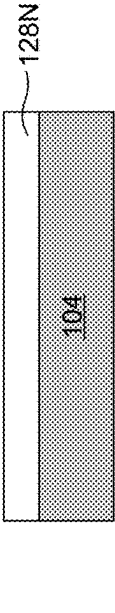
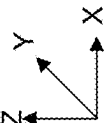


Fig. 7B



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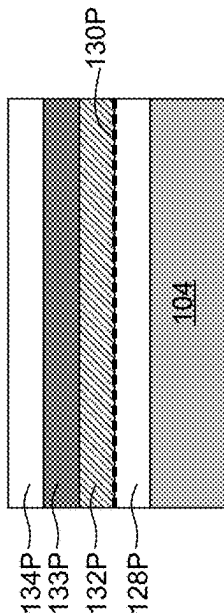


Fig. 8A

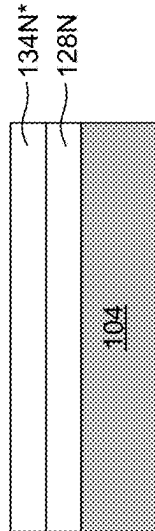
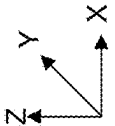
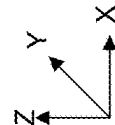


Fig. 8B

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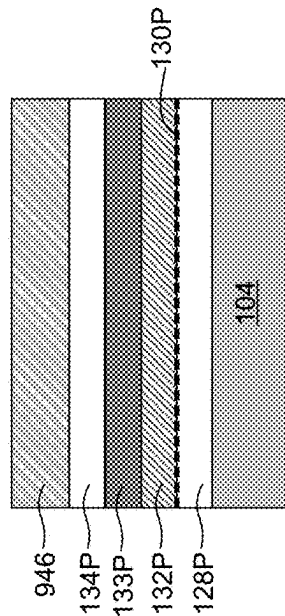
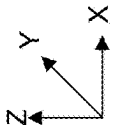


Fig. 9A



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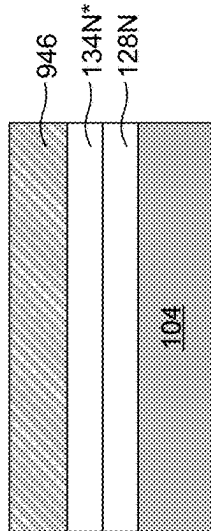
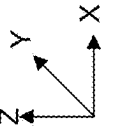
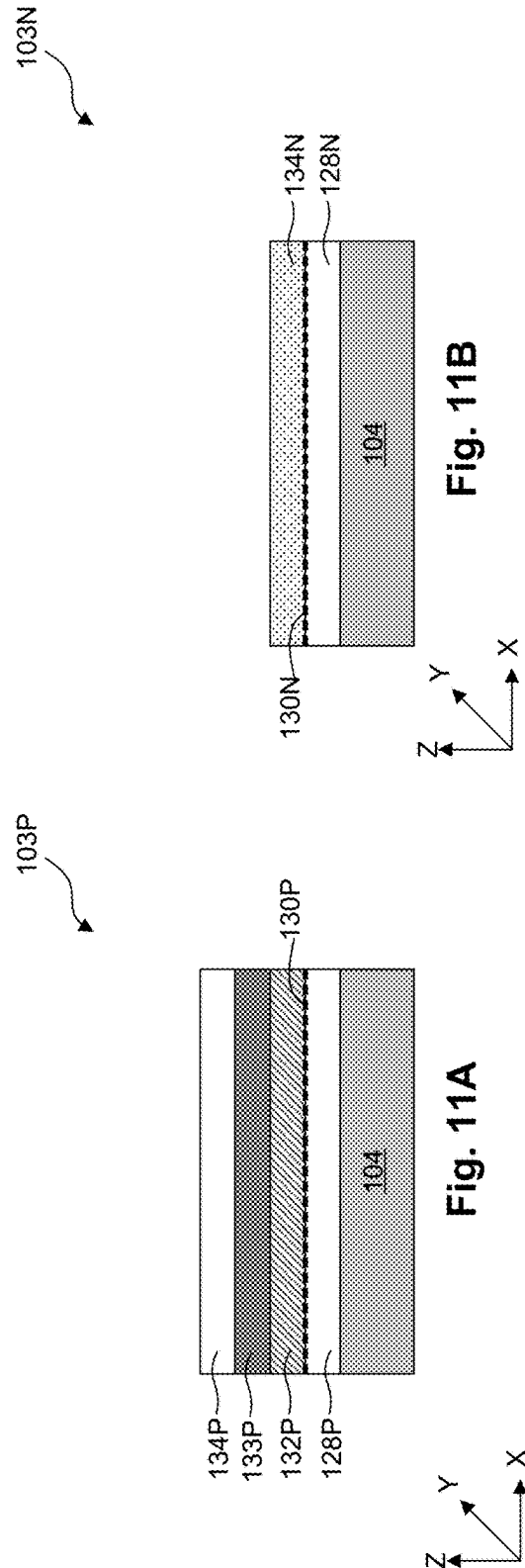
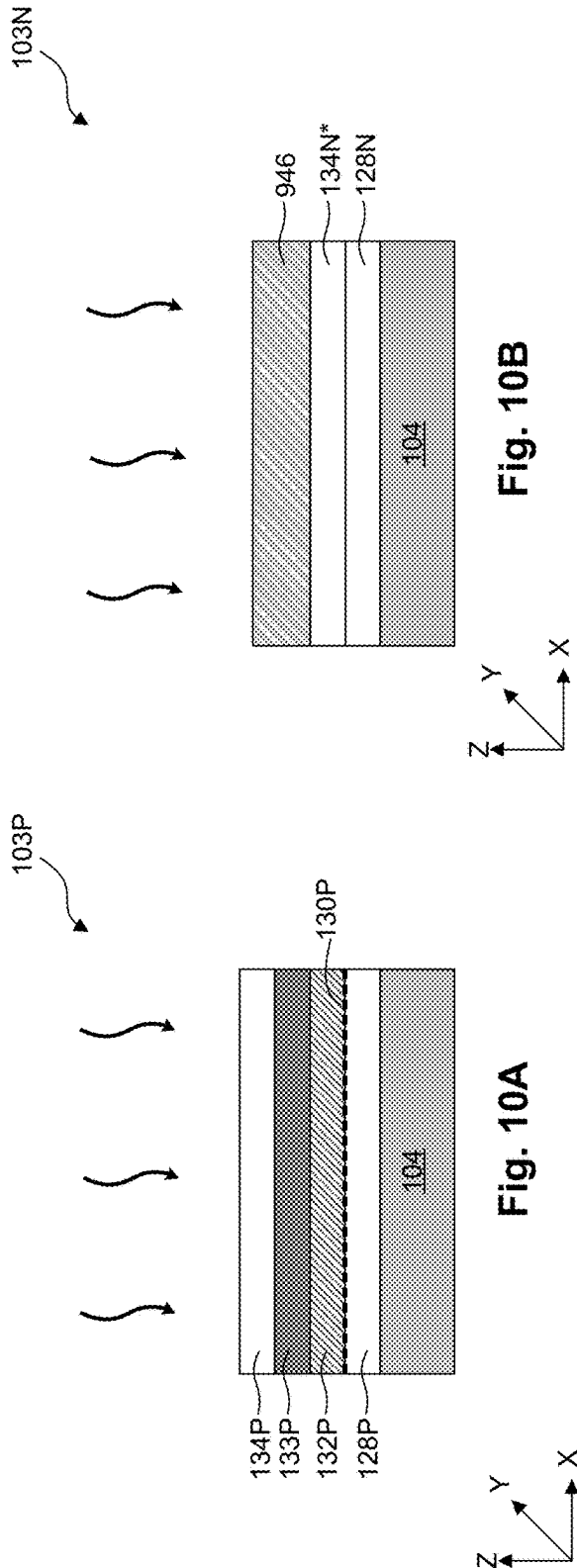
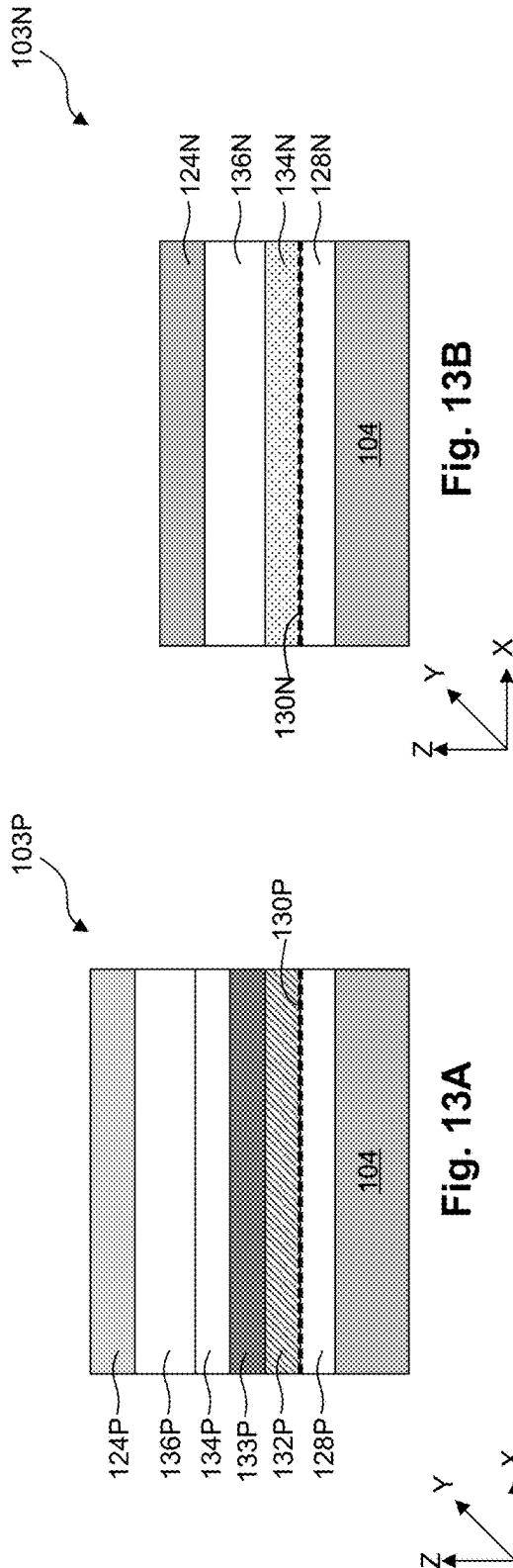
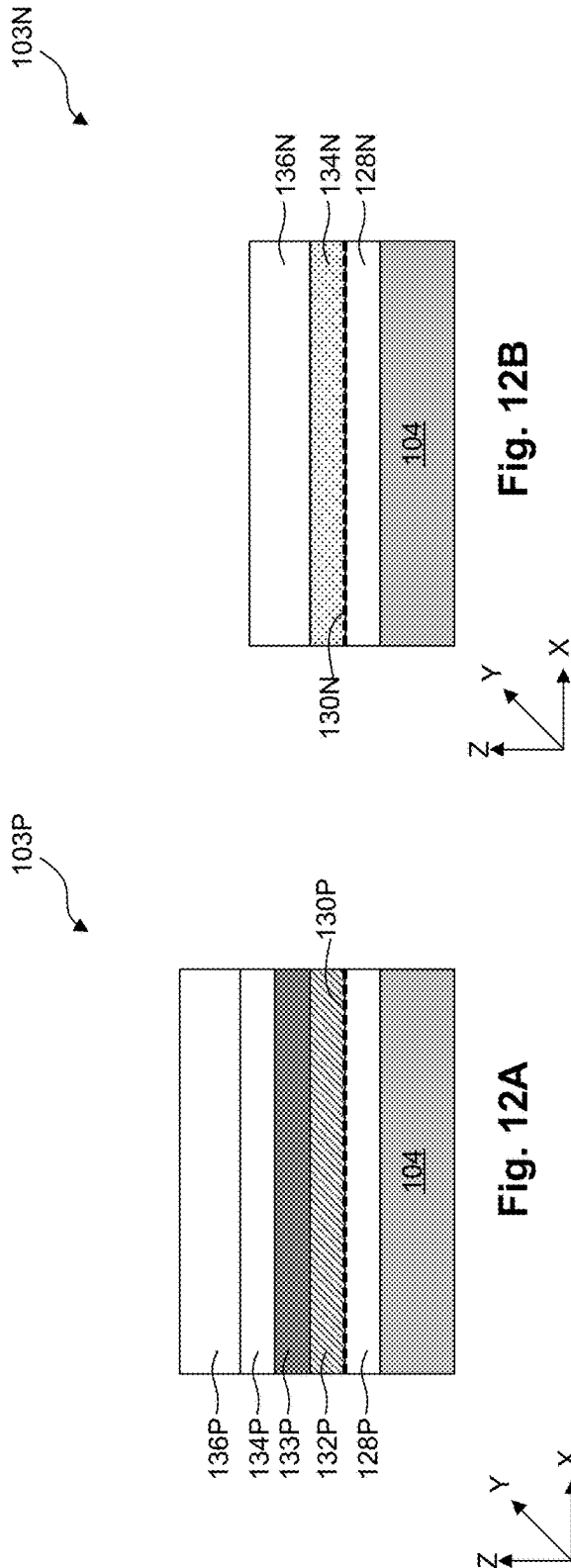


Fig. 9B







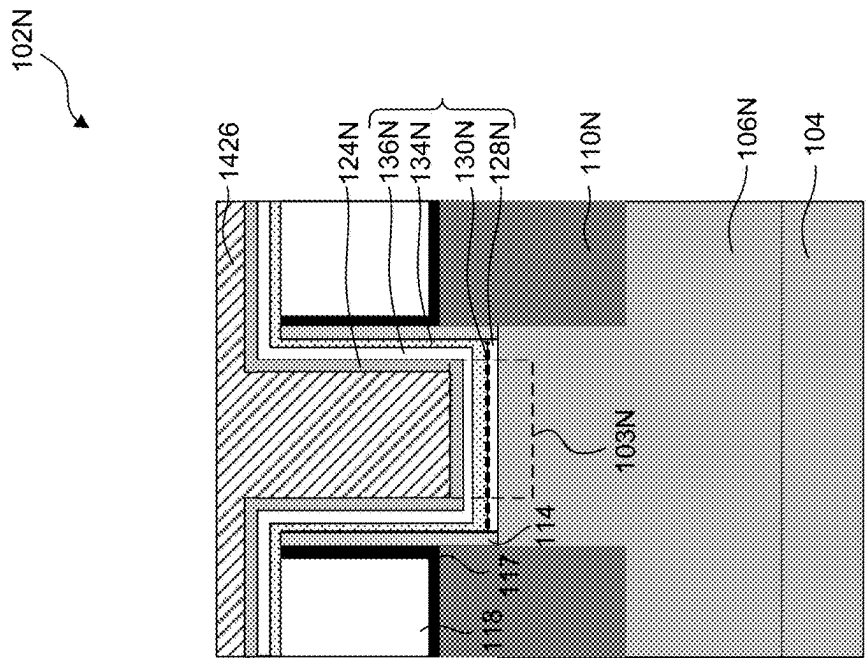


Fig. 14B

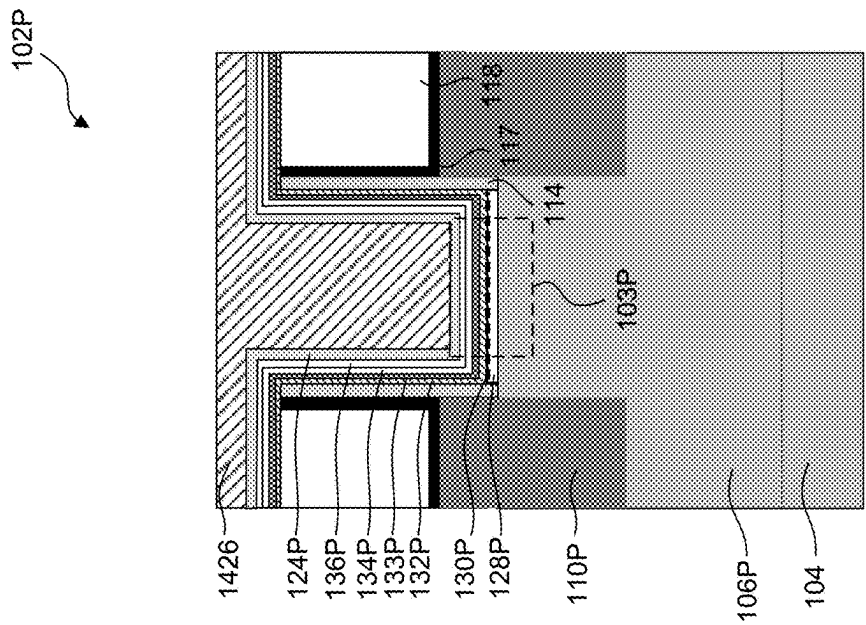
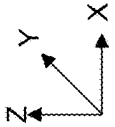
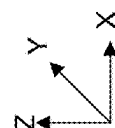


Fig. 14A



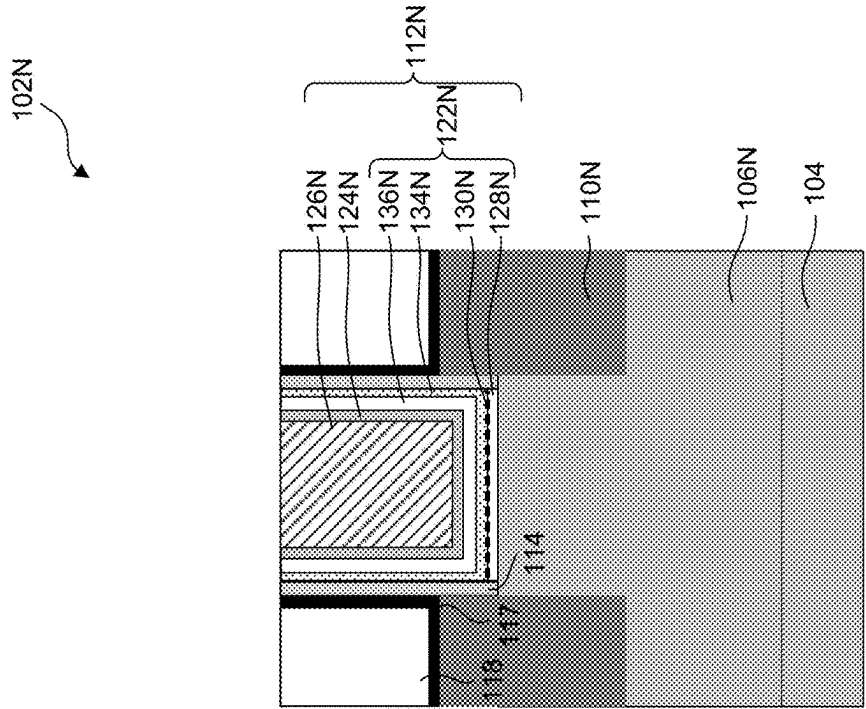


Fig. 15B

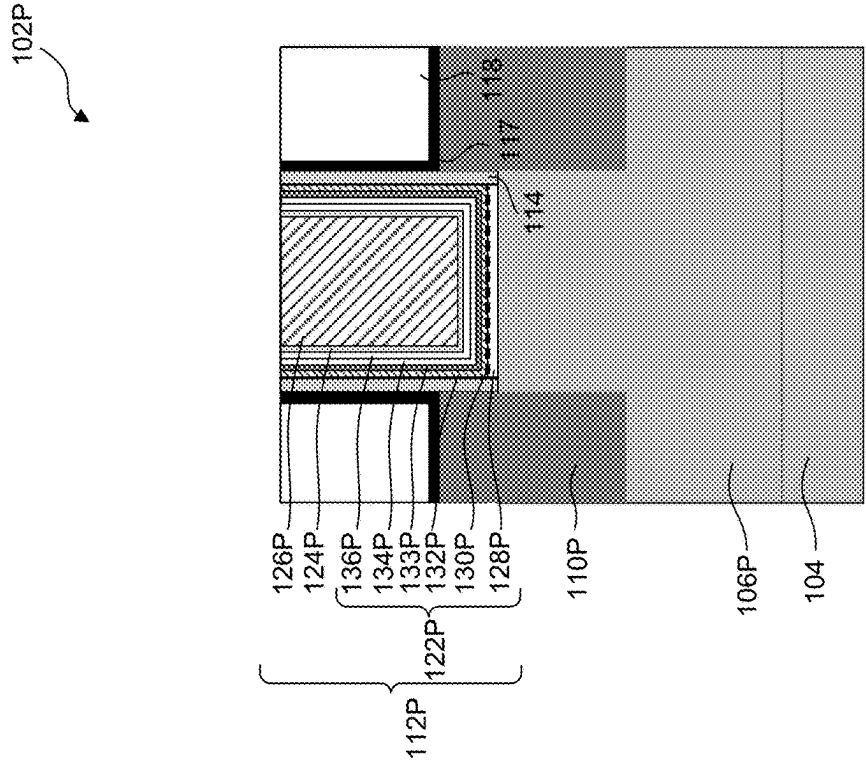
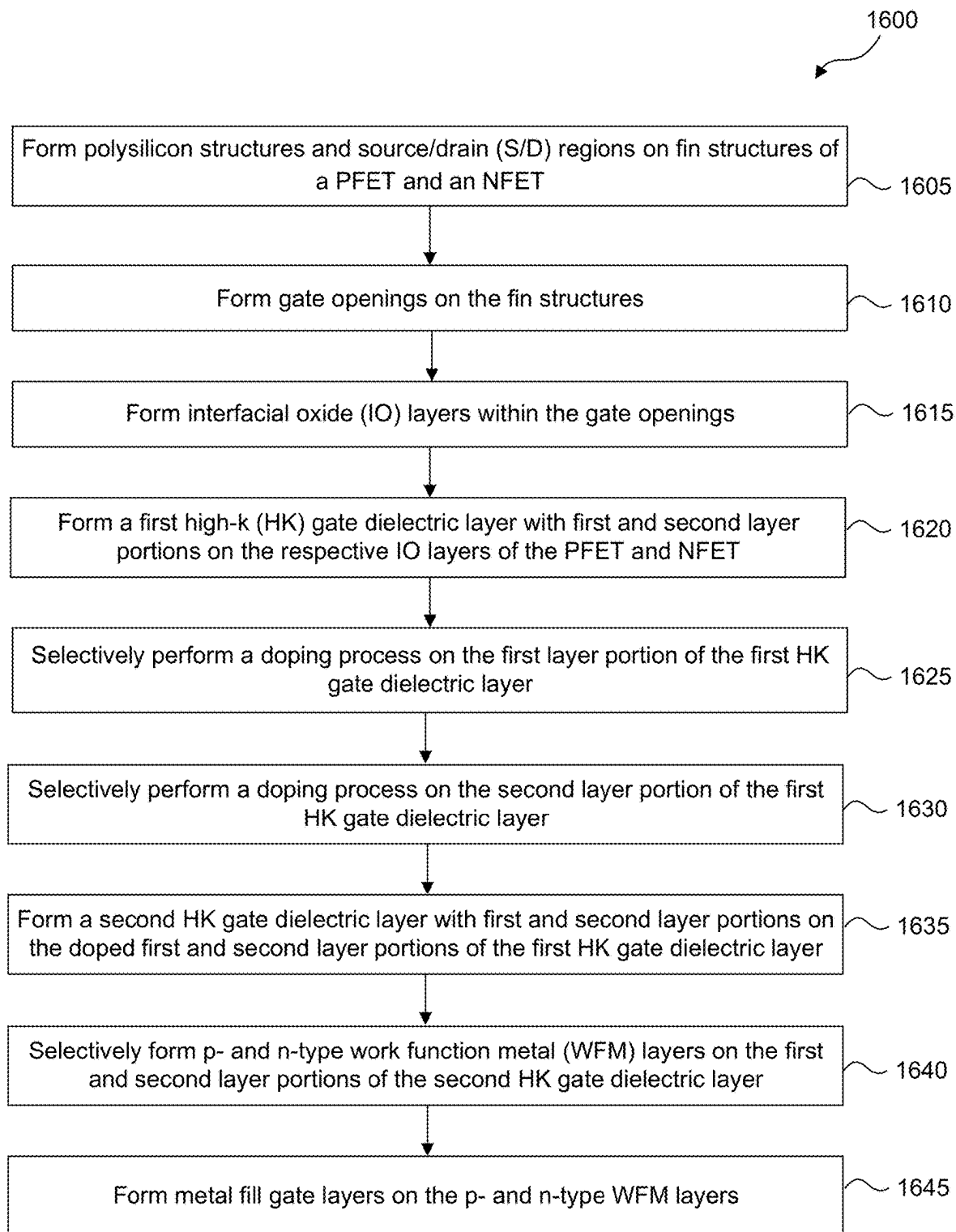


Fig. 15A

**Fig. 16**

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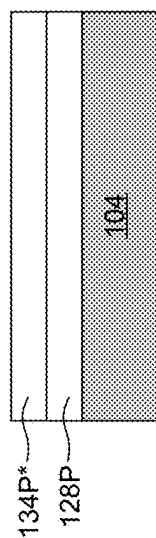
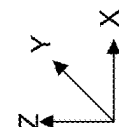


Fig. 17A



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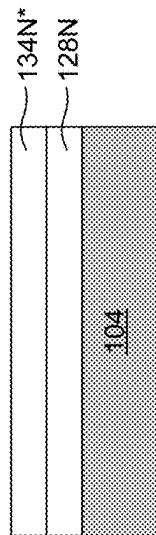
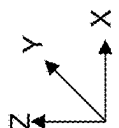


Fig. 17B



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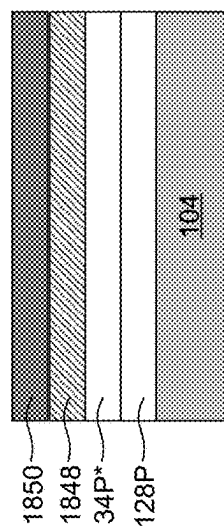
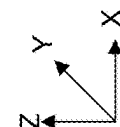


Fig. 18A



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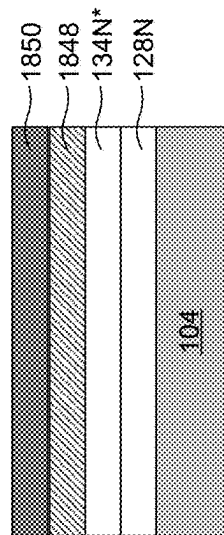
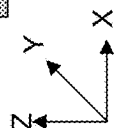


Fig. 18B



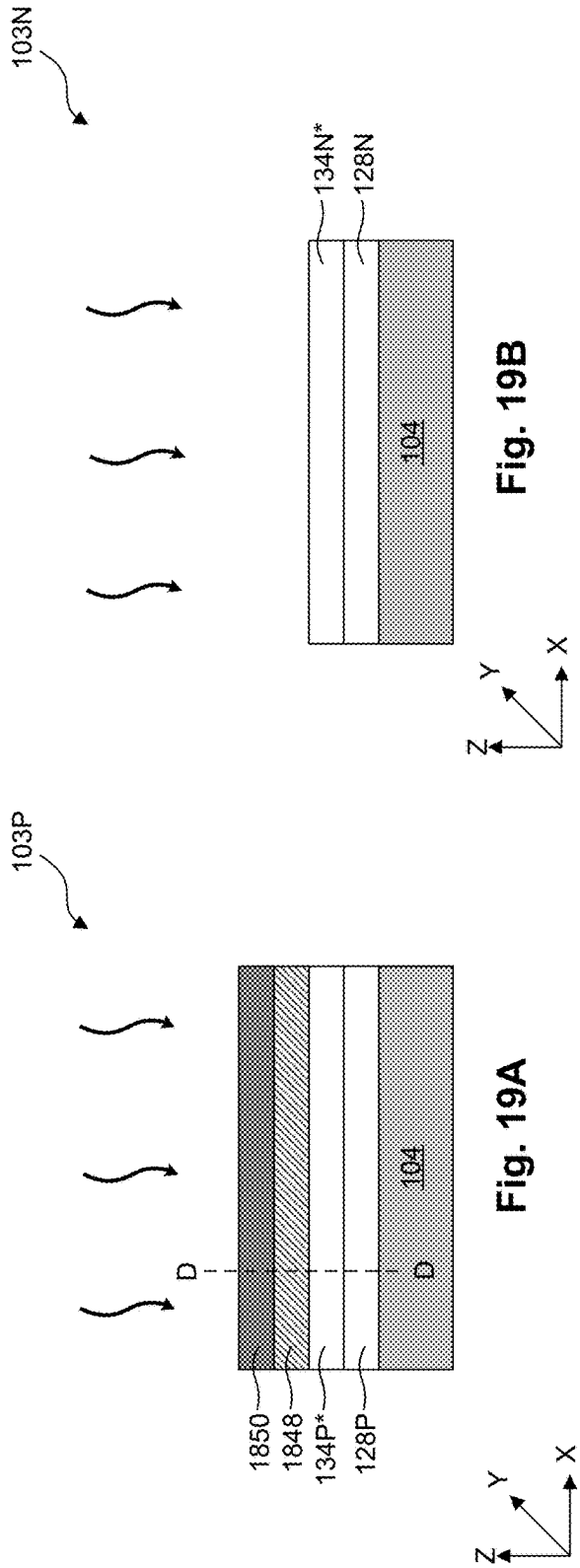


Fig. 19B

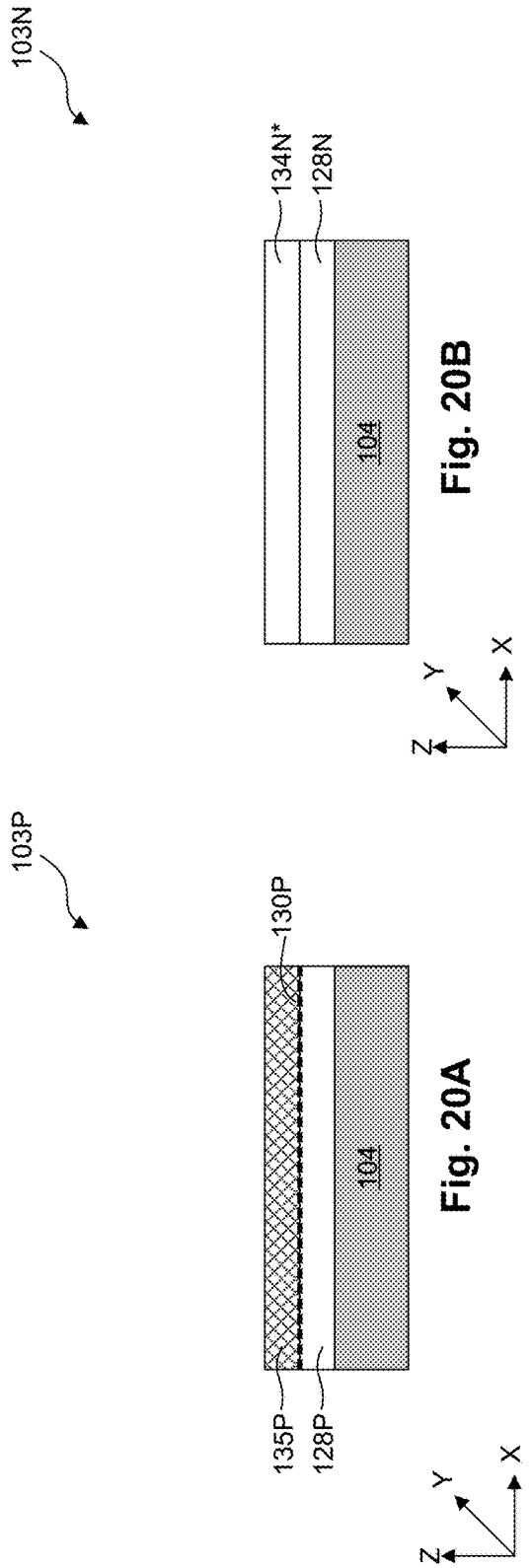
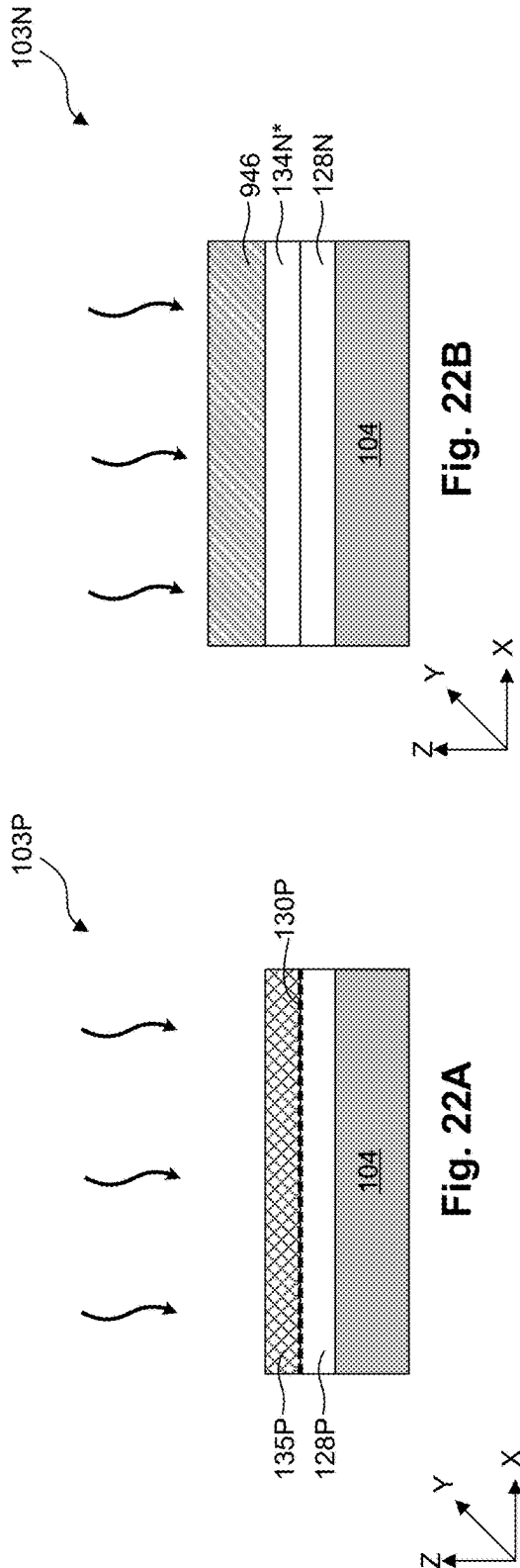
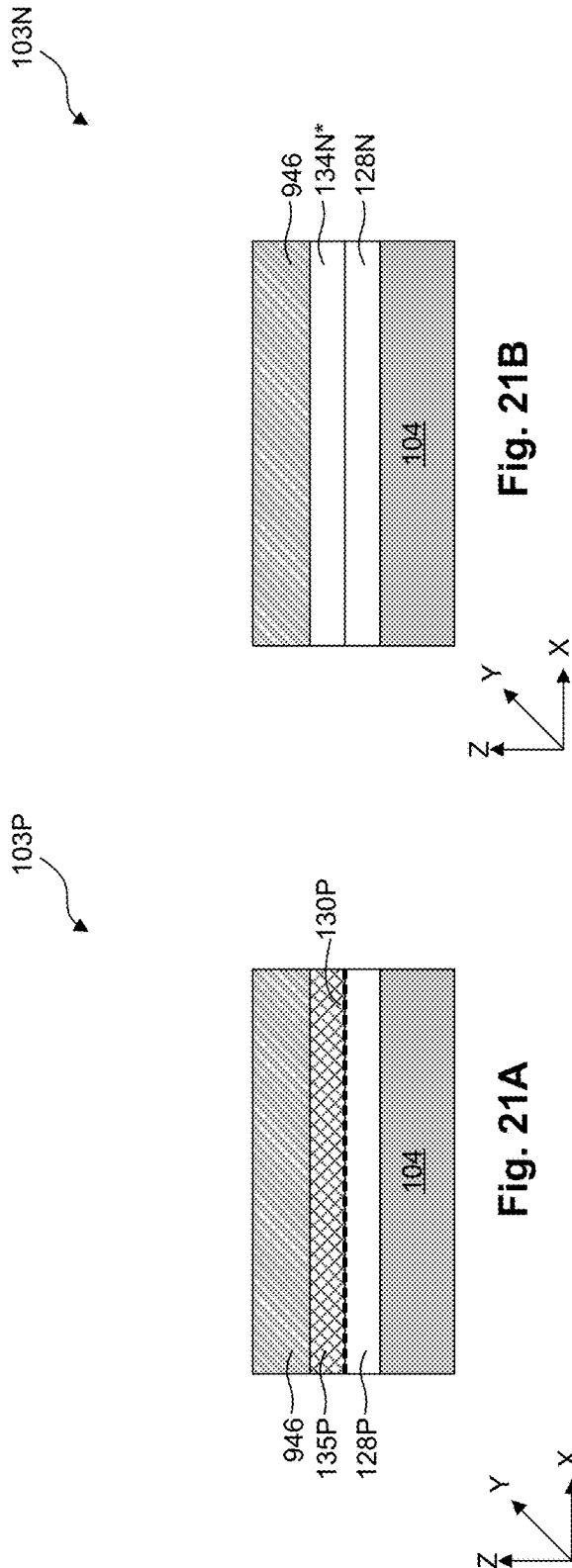
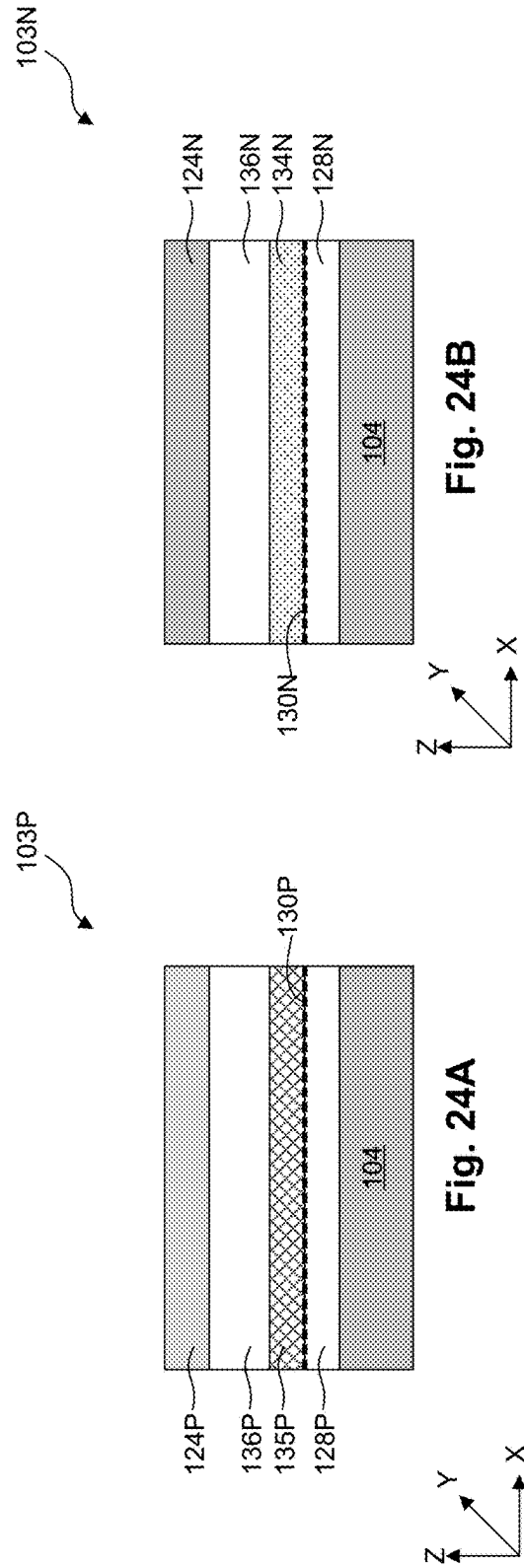
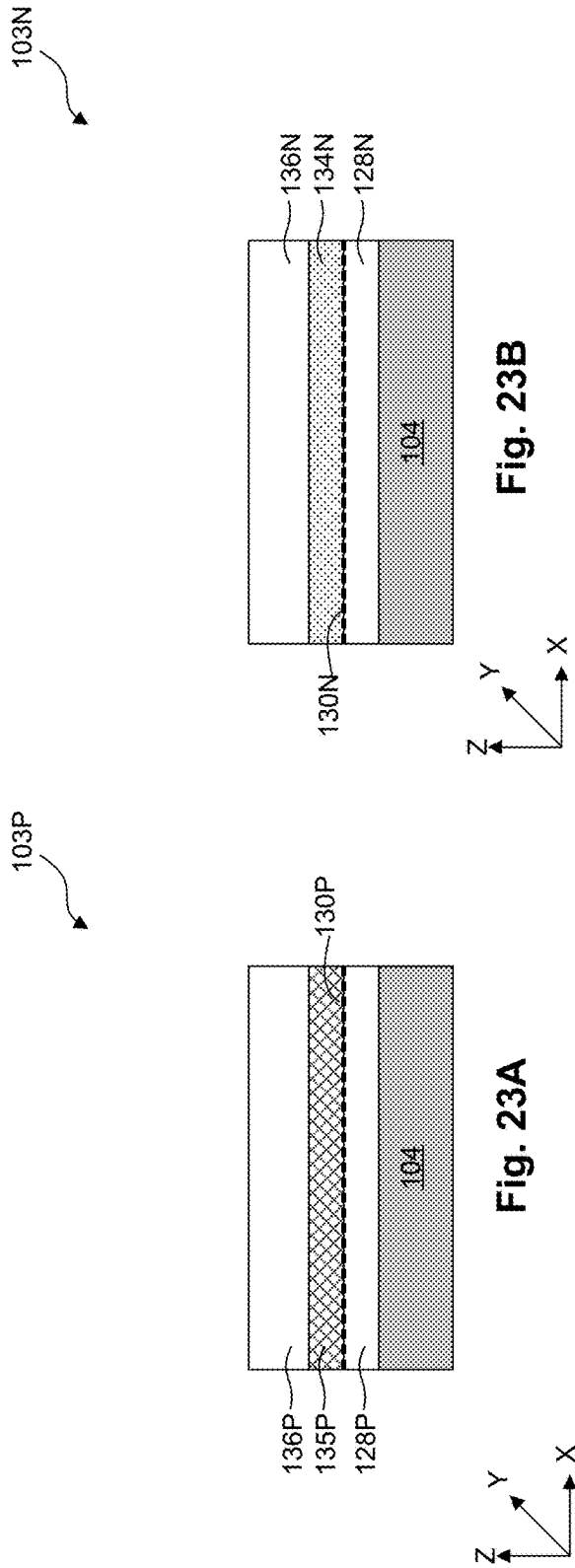


Fig. 20B





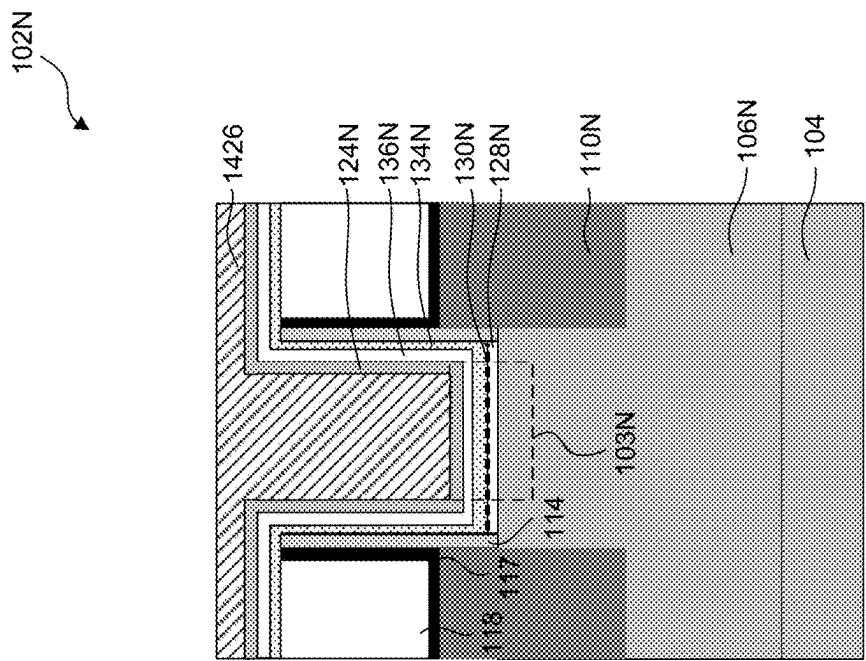


Fig. 25B

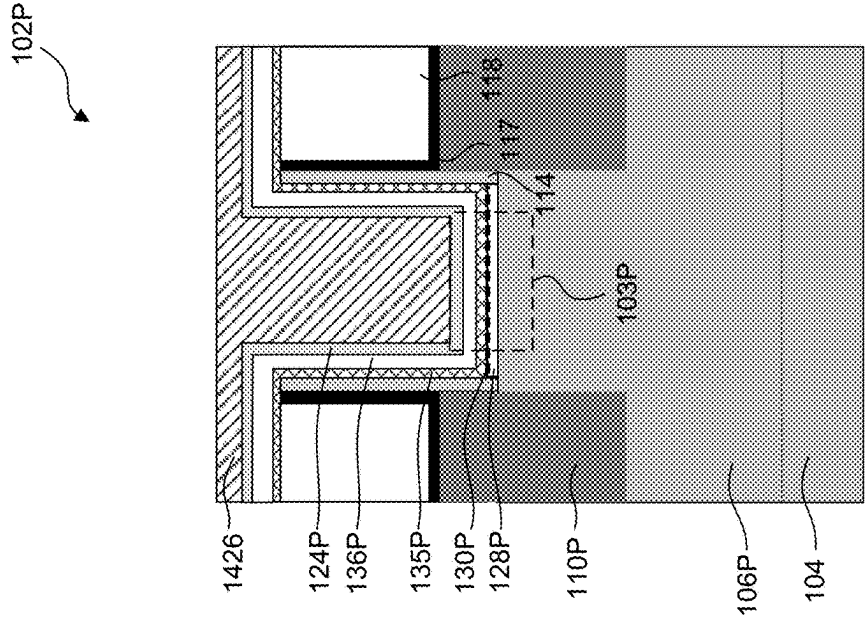


Fig. 25A

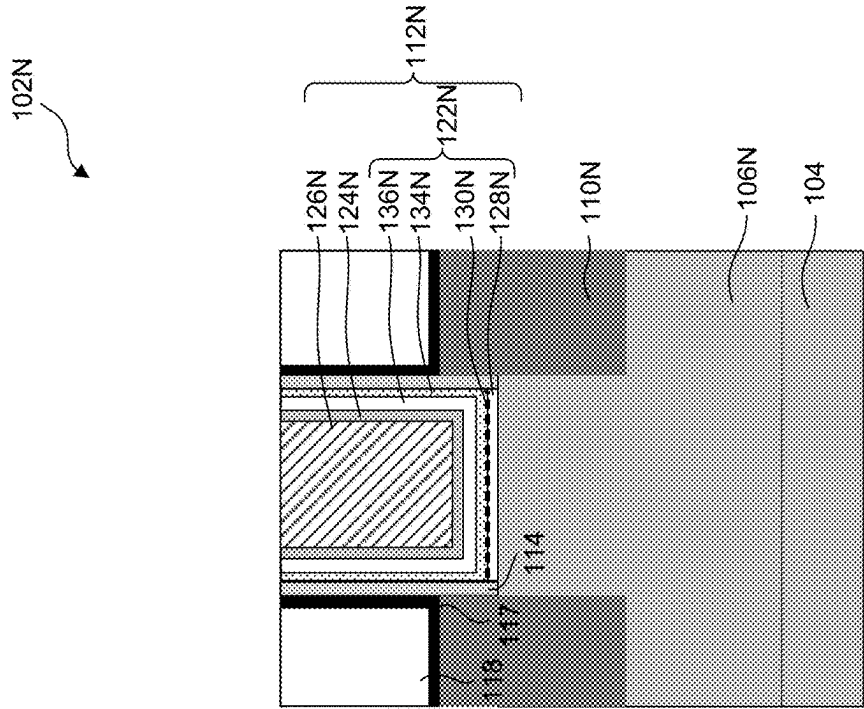


Fig. 26B

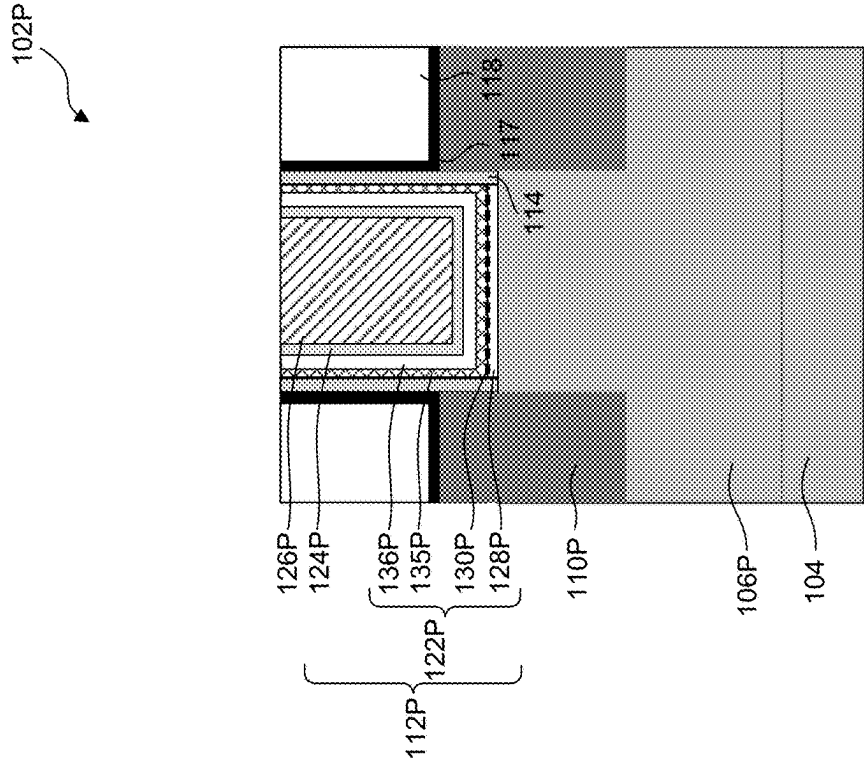
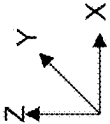
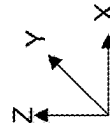
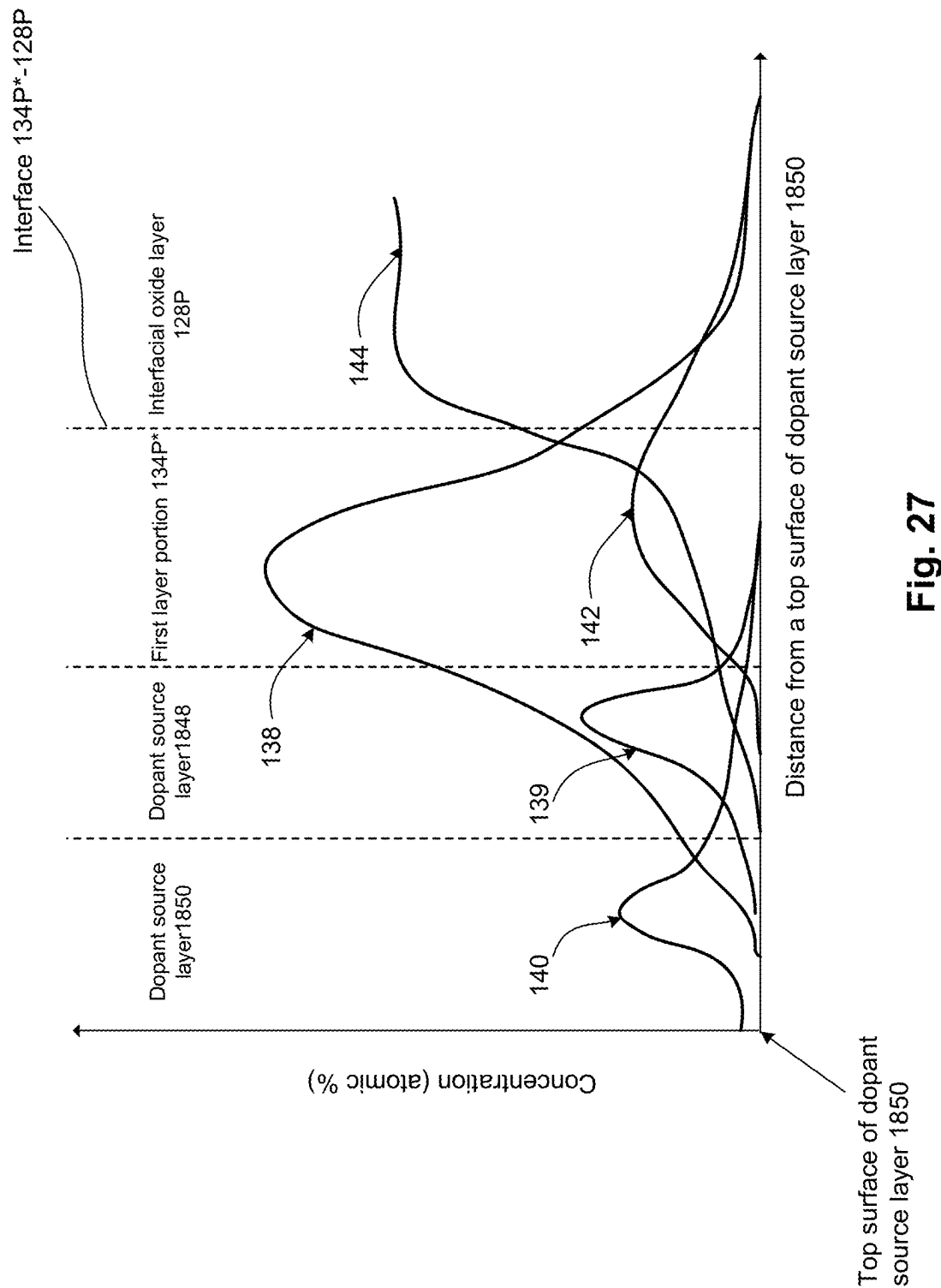


Fig. 26A





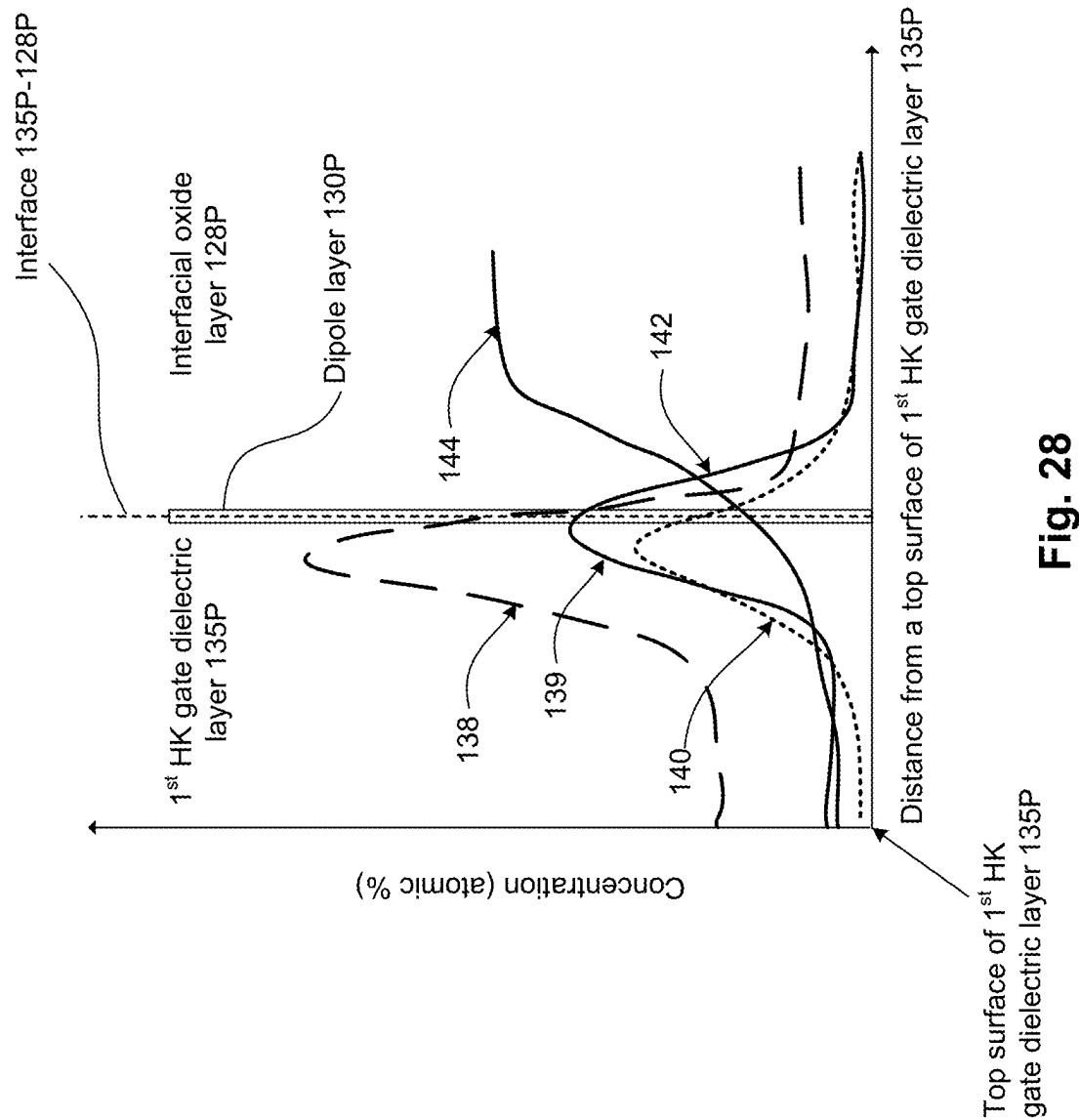


Fig. 28

GATE STRUCTURES IN SEMICONDUCTOR DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/406,879, titled “Gate Structures in Semiconductor Devices,” filed Aug. 19, 2021, which is incorporated by reference in its entirety.

BACKGROUND

With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs, fin field effect transistors (finFETs), and gate-all-around (GAA) FETs. Such scaling down has increased the complexity of semiconductor manufacturing processes.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures.

FIG. 1A illustrates an isometric view of a semiconductor device, in accordance with some embodiments.

FIGS. 1B-1E illustrate cross-sectional views of a semiconductor device with different gate structures, in accordance with some embodiments.

FIG. 1F illustrates device characteristics of a semiconductor device with different gate structures, in accordance with some embodiments.

FIG. 2 is a flow diagram of a method for fabricating a semiconductor device with different gate structures, in accordance with some embodiments.

FIGS. 3A-15B illustrate cross-sectional views of a semiconductor device with different gate structures at various stages of its fabrication process, in accordance with some embodiments.

FIG. 16 is a flow diagram of a method for fabricating a semiconductor device with different gate structures, in accordance with some embodiments.

FIGS. 17A-26B illustrate cross-sectional views of a semiconductor device with different gate structures at various stages of its fabrication process, in accordance with some embodiments.

FIGS. 27-28 illustrates device characteristics of a semiconductor device with different gate structures at various stages of its fabrication process, in accordance with some embodiments.

Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely

examples and are not intended to be limiting. For example, the process for forming a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

The fin structures disclosed herein may be patterned by any suitable method. For example, the fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes can combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fin structures.

The required gate voltage—the threshold voltage (V_t)—to turn on a field effect transistor (FET) can depend on the

semiconductor material of the FET channel region and/or the effective work function (EWF) value of a gate structure of the FET. For example, for an n-type FET (NFET), reducing the difference between the EWF value(s) of the NFET gate structure and the conduction band energy of the material (e.g., 4.1 eV for Si or 3.8 eV for SiGe) of the NFET channel region can reduce the NFET threshold voltage. For a p-type FET (PFET), reducing the difference between the EWF value(s) of the PFET gate structure and the valence band energy of the material (e.g., 5.2 eV for Si or 4.8 eV for SiGe) of the PFET channel region can reduce the PFET threshold voltage. The EWF values of the FET gate structures can depend on the thickness and/or material composition of each of the layers of the FET gate structure. As such, FETs can be manufactured with different threshold voltages by adjusting the thickness and/or material composition of the FET gate structures.

Due to the increasing demand for multi-functional low power portable devices, there is an increasing demand for FETs with lower and/or different threshold voltages, such as threshold voltages lower than 100 mV. One way to achieve multi-Vt devices with low threshold voltages in FETs can be with different work function metal (WFM) layer thicknesses greater than about 4 nm (e.g., about 5 nm to about 10 nm) in the gate structures. However, the different WFM layer thicknesses can be constrained by the FET gate structure geometries. Also, depositing different WFM layer thicknesses can become increasingly challenging with the continuous scaling down of FETs (e.g., GAA FETs, finFETs, and/or MOSFETs).

The present disclosure provides example multi-Vt devices with FETs (e.g., finFETs) having ultra-low threshold voltages (e.g., about 20 mV to about 100 mV) different from each other and provides example methods of forming such FETs on the same substrate. The example methods form NFETs and PFETs with WFM layer of similar thicknesses, but with ultra-low and/or different threshold voltages, on the same substrate. These example methods can be more cost-effective (e.g., cost reduced by about 20% to about 30%) and time-efficient (e.g., time reduced by about 15% to about 20%) in manufacturing reliable FET gate structures with lower and/or different threshold voltages than other methods of forming FETs with similar dimensions and threshold voltages on the same substrate. In addition, these example methods can form FET gate structures with much smaller dimensions (e.g., thinner gate stacks) than other methods of forming FETs with similar threshold voltages.

In some embodiments, NFETs and PFETs with different gate structure configurations, but with similar WFM layer thicknesses, can be selectively formed on the same substrate to achieve ultra-low and/or different threshold voltages. The different gate structures can have high-K (HK) gate dielectric layers doped with different metallic dopants. The different metal dopants can induce dipoles of different polarities and/or concentrations at interfaces between the HK gate dielectric layers and interfacial oxide (IO) layers. The dipoles of different polarities and/or concentrations result in gate structures with different EWF values and threshold voltages. In some embodiments, HK gate dielectric layers can be doped with dopants of two different materials to induce dipoles of different materials that provide stronger electric fields and lower threshold voltages. Thus, controlling the dopant materials and/or concentrations in the HK gate dielectric layers can tune the EWF values of the NFET and PFET gate structures, and as a result can adjust the threshold voltages of the NFETs and PFETs without varying the WFM layer thicknesses. In some embodiments, instead

of the doped HK gate dielectric layer, PFET gate structure can include dual metal oxide layers interposed between the HK gate dielectric and the IO layer to induce dipoles of different materials between the HK gate dielectric layer and the IO layer.

FIG. 1A illustrates an isometric view of a semiconductor device **100** with PFET **102P** and NFET **102N**, according to some embodiments. PFET **102P** and NFET **102N** can have different cross-sectional views, as illustrated in FIGS. 1B-1E, according to various embodiments. FIGS. 1B-1E illustrate cross-sectional views of PFET **102P** and NFET **102N** along respective lines A-A and B-B of FIG. 1A. FIGS. 1B-1E illustrate cross-sectional views of semiconductor device **100** with additional structures that are not shown in FIG. 1A for simplicity. The discussion of elements of PFET **102P** and NFET **102N** with the same annotations applies to each other, unless mentioned otherwise.

Referring to FIG. 1A, NFET **102N** can include an array of gate structures **112N** disposed on fin structure **106N**, and PFET **102P** can include an array of gate structures **112P** disposed on fin structure **106P**. NFET **102N** can further include an array of S/D regions **110N** disposed on portions of fin structure **106N** that are not covered by gate structures **112N**. Similarly, PFET **102P** can further include an array of epitaxial S/D regions **110P** disposed on portions of fin structure **106P** that are not covered by gate structures **112P**.

Semiconductor device **100** can further include gate spacers **114**, shallow trench isolation (STI) regions **116**, etch stop layers (ESLs) **117**, and interlayer dielectric (ILD) layers **118**. In some embodiments, gate spacers **114**, STI regions **116**, ESLs **117**, and ILD layers **118** can include an insulating material, such as silicon oxide, silicon nitride (SiN), silicon carbon nitride (SiCN), silicon oxycarbon nitride (SiOCN), and silicon germanium oxide. In some embodiments, gate spacers **114** can have a thickness of about 2 nm to about 9 nm for adequate electrical isolation of gate structures **112N** and **112P** from adjacent structures.

Semiconductor device **100** can be formed on a substrate **104** with PFET **102P** and NFET **102N** formed on different regions of substrate **104**. There may be other FETs and/or structures (e.g., isolation structures) formed between PFET **102P** and NFET **102N** on substrate **104**. Substrate **104** can be a semiconductor material, such as silicon, germanium (Ge), silicon germanium (SiGe), a silicon-on-insulator (SOI) structure, and a combination thereof. Further, substrate **104** can be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic). In some embodiments, fin structures **106P-106N** can include a material similar to substrate **104** and extend along an X-axis.

Referring to FIGS. 1B-1C, PFET-NFET **102P-102N** can include S/D regions **110P-110N** and gate structures **112P-112N**. For PFET **102P**, S/D regions **110P** can include an epitaxially-grown semiconductor material, such as Si and SiGe, and p-type dopants, such as boron and other suitable p-type dopants. For NFET **102N**, S/D regions **110N** can include an epitaxially-grown semiconductor material, such as Si, and n-type dopants, such as phosphorus and other suitable n-type dopants.

Gate structures **112P-112N** can be multi-layered structures. Gate structures **112P-112N** can include (i) gate oxide structures **122P-122N** disposed on respective fin structures **106P-106N**, (ii) work function metal (WFM) layers **124P-124N** disposed on respective gate oxide structures **122P-122N**, and (iii) gate metal fill layers **126P-126N** disposed on respective WFM layers **124P-124N**.

Referring to FIG. 1B, in some embodiments, gate oxide structure **122P** can include (i) an IO layer **128P** disposed on fin structure **106P**, (ii) a first metal oxide layer **132P** disposed on IO layer **128P**, (iii) a second metal oxide layer **133P** disposed on first metal oxide layer **133P**, (iv) a dipole layer **130P** disposed at an interface between IO layer **128P** and first metal oxide layer **132P**, (v) a first HK gate dielectric layer **134P** disposed on second metal oxide layer **133P**, and (vi) a second HK gate dielectric layer **136P** disposed on first HK gate dielectric layer **134P**.

IO layer **128P** can include an oxide of the material of fin structure **106P**, such as silicon oxide (SiO_2), silicon germanium oxide (SiGeO_x), and germanium oxide (GeO_x). The materials of first and second metal oxide layers **132P-133P** are different from each other and can induce the formation of two different p-type dipoles in dipole layer **130P**. Dipole layer **130P** can include p-type dipoles of (i) first metal ions from first metal oxide layer **132P** and oxygen ions from IO layer **128P** and (ii) second metal ions from second metal oxide layer **133P** and oxygen ions from IO layer **128P**. The first metal ions are different from the second metal ions. First and second metal oxide layers **132P-133P** can include oxides of metals that have electronegativity values greater than the electronegativity values of metals or semiconductors in first HK gate dielectric layer **134P**. In addition, first and second metal oxide layers **132P-133P** can include oxide materials that have oxygen areal densities greater than the oxygen areal densities of oxide materials in first HK gate dielectric layer **134P**. As used herein, the term "oxygen areal density" of an oxide material refers to an atomic concentration of oxygen atoms per unit area of the oxide material.

The larger electronegativity values and oxygen areal densities of first and second metal oxide layers **132P-133P** can induce stronger p-type dipoles in dipole layer **130P** compared to dipoles induced at an interface between IO layer **128P** and first HK gate dielectric layer **134P** in the absence of first and second metal oxide layers **132P-133P**. Also, the two different p-type dipoles can generate a stronger electric field than single p-type dipoles and form a more stable dipole layer **130P**. As stronger p-type dipoles can result in lower threshold voltages for PFETs, the use of first and second metal oxide layers **132P-133P** can form PFET **102P** with a threshold voltage lower than about 100 mV (e.g., about 50 mV, about 30 mV, or about 20 mV).

In some embodiments, first and second metal oxide layers **132P-133P** can include oxides of transition metals, such as zinc oxide (ZnO_2), niobium oxide (NbO_2), molybdenum oxide (MoO_2), tungsten oxide (WO_3), and tantalum oxide (Ta_2O_5). In some embodiments, first and second metal oxide layers **132P-133P** can include oxides of elements from group 13 of the periodic table, such as gallium oxide (Ga_2O_3), aluminum oxide (Al_2O_3), and indium oxide (In_2O_3), when first HK gate dielectric layer **134P** includes HfO_2 . In some embodiments, first metal oxide layer **132P** can include an oxide of a transition metal and second metal oxide layer **133P** can include an oxide of a material from group 13 of the periodic table. In some embodiments, dipole layer **130P** can include Ga—O and Zn—O dipoles when first metal oxide layer **132P** includes ZnO and second metal oxide layer **133P** includes Ga_2O_3 .

In some embodiments, dipole layer **130P** can have a higher concentration of transition metal based dipoles (e.g., Zn—O, Nb—O, Mo—O, W—O, or Ta—O) than group 13 element based dipoles (e.g., Ga—O, Al—O, or In—O) to achieve threshold voltages below about 50 mV (e.g., about 30 mV or 20 mV). In such embodiments, first metal oxide layer **132P** includes a transition metal oxide with a thickness

greater than the thickness of second metal oxide **133P** that includes an oxide of group 13 element as the concentration of dipoles is directly proportional to the thickness of the dipole source layer. In contrast, to achieve threshold voltages greater than 50 mV (e.g. about 70 mV or about 100 mV), dipole layer **130P** can have a higher concentration of group 13 element based dipoles than transition metal based dipoles. In such embodiments, first metal oxide layer **132P** includes an oxide of group 13 element with a thickness greater than the thickness of second metal oxide **133P** that includes a transition metal oxide. In some embodiments, first metal oxide layer **132P** can include oxides of transition metals and dipole layer **130P** can include transition metal based dipoles in the absence of second metal oxide layer **133P**.

In some embodiments, first and second metal oxide layers **132P-133P** can have thicknesses ranging from about 0.5 nm to about 3 nm. If first and second metal oxide layers **132P-133P** are thinner than about 0.5 nm, the formation of dipoles in dipole layer **130P** may not occur. On the other hand, if first and second metal oxide layers **132P-133P** are thicker than 3 nm, diffusion of metal atoms from first and second metal oxide layers **132P-133P** may degrade first and second HK gate dielectric layers **134P-136P**, and consequently degrade device performance.

First and second HK gate dielectric layers **134P-136P** can include high-k dielectric materials, such as hafnium oxide (HfO_2), titanium oxide (TiO_2), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta_2O_3), hafnium silicate (Hf-SiO_4), zirconium oxide (ZrO_2), and zirconium silicate (Zr-SiO_2). In some embodiments, first and second HK gate dielectric layers **134P-136P** can include materials similar to or different from each other. In some embodiments, first and second HK gate dielectric layers **134P-136P** can have thicknesses similar to or different from each other. In some embodiments, first and second HK gate dielectric layers **134P-136P** can be undoped.

In some embodiments, p-type WFM (pWFM) layer **124P** can include substantially Al-free (e.g., with no Al) (i) Ti-based nitrides or alloys, such as TiN, TiSiN, titanium gold (Ti—Au) alloy, titanium copper (Ti—Cu) alloy, titanium chromium (Ti—Cr) alloy, titanium cobalt (Ti—Co) alloy, titanium molybdenum (Ti—Mo) alloy, and titanium nickel (Ti—Ni) alloy; (ii) Ta-based nitrides or alloys, such as TaN, TaSiN, Ta—Au alloy, Ta—Cu alloy, Ta—W alloy, tantalum platinum (Ta—Pt) alloy, Ta—Mo alloy, Ta—Ti alloy, and Ta—Ni alloy; or (iii) a combination thereof. In some embodiments, pWFM layer **124P** can include a thickness ranging from about 1 nm to about 3 nm. Other suitable dimensions of pWFM layer **124P** are within the scope of the present disclosure. Gate metal fill layer **126P** can include a suitable conductive material, such as tungsten (W), Ti, silver (Ag), ruthenium (Ru), molybdenum (Mo), copper (Cu), cobalt (Co), Al, iridium (Ir), nickel (Ni), metal alloys, and a combination thereof.

Referring to FIG. 1C, in some embodiments, gate oxide structure **122N** can include (i) an IO layer **128N** disposed on fin structure **106N**, (ii) a first HK gate dielectric layer **134N** disposed on IO layer **128N**, (iii) a second HK gate dielectric layer **136N** disposed on first HK gate dielectric layer **134N**, and (iv) a dipole layer **130N** disposed at an interface between IO layer **128P** and first HK gate dielectric layer **134N**.

IO layer **128N** can include an oxide of the material of fin structure **106N**, such as silicon oxide (SiO_2), silicon germanium oxide (SiGeO_x), and germanium oxide (GeO_x). In some embodiments, first HK gate dielectric layer **134N** can

include dopants of metals that have electronegativity values lower than the electronegativity values of metallic or semiconductor materials included in first HK gate dielectric layer **134N**. In some embodiments, first HK gate dielectric layer **134N** can include dopants of a rare-earth metal, such as Lanthanum (La), Yttrium (Y), Scandium (Sc), Cerium (Ce), Ytterbium (Yb), Erbium (Er), Dysprosium (Dy), and Lutetium (Lu). The metal dopants of first HK gate dielectric layer **134N** can induce the formation of n-type dipoles in dipole layer **130N**. Dipole layer **130N** can include n-type dipoles of metal ions from the metal dopants and oxygen ions from IO layer **128N**, such as La—O dipoles, when first HK gate dielectric layer **134N** includes La dopants. The lower electronegativity value of the metal dopants of first HK gate dielectric layer **134N** can induce stronger n-type dipoles in dipole layer **130N** compared to dipoles induced at an interface between IO layer **128P** and undoped first HK gate dielectric layer **134N**. In some embodiments, first and second HK gate dielectric layers **134N-136N** can include high-k dielectric materials similar to first and second HK gate dielectric layers **134P-136P**.

In some embodiments, nWFM layer **124N** can include titanium aluminum (TiAl), titanium aluminum carbide (TiAlC), tantalum aluminum (TaAl), tantalum aluminum carbide (TaAlC), Al-doped Ti, Al-doped TiN, Al-doped Ta, Al-doped TaN, or a combination thereof. In some embodiments, nWFM layer **124N** can include a thickness ranging from about 1 nm to about 3 nm. Other suitable dimensions of nWFM layer **124N** are within the scope of the present disclosure. Gate metal fill layer **126N** can include a conductive material similar to gate metal fill layer **126P**.

Referring to FIG. 1D, in some embodiments, gate oxide structure **122P** does not include first and second metal oxide layers **132P-133P** and includes first HK gate dielectric layer **135P** with dual-metal dopants instead of undoped first HK gate dielectric layer **134P** (shown in FIG. 1B). Instead of first and second metal oxide layers **132P-133P**, the dual-metal dopants of first HK gate dielectric layer **135P** induces the formation of different p-type dipoles in dipole layer **130P**. Dipole layer **130P** can include p-type dipoles of two different metal ions from the dual-metal dopants of first HK gate dielectric layer **135P** and oxygen ions from IO layer **128P**. First HK gate dielectric layer **135P** can include dopants of metals that have electronegativity values greater than the electronegativity values of metals or semiconductors in first HK gate dielectric layer **135P**. In some embodiments, first HK gate dielectric layer **135P** can include dopants of materials from group 13 of the periodic table, such as Ga, Al, and In, when first HK gate dielectric layer **135P** includes HfO₂. In some embodiments, first HK gate dielectric layer **135P** can include dopants of transition metals, such as Zn, Nb, Mo, W, and Ta. In some embodiments, dipole layer **130P** can include Ga—O and Zn—O dipoles when first HK gate dielectric layer **135P** includes Ga and Zn.

In some embodiments, first HK gate dielectric layer **135P** can have a higher concentration of transition metal based dopants (e.g., Zn, Nb, Mo, W, or Ta) than group 13 element based dipoles (e.g., Ga, Al, or In) to achieve threshold voltages below about 50 mV (e.g., about 30 mV or 20 mV) as the concentration of dopants is directly proportional to the concentration of dipoles. In contrast, to achieve threshold voltages greater than 50 mV (e.g., about 70 mV or about 100 mV), first HK gate dielectric layer **135P** can have a higher concentration of group 13 element based dopants than transition metal based dopants. In some embodiments, first

HK gate dielectric layer **135P** can include transition metal based dopants and may not include group 13 element based dopants.

FIG. 1F shows the Hf, Zn, Ga, O₂, and Si concentration profiles **138**, **139**, **140**, **142**, and **144** across first HK gate dielectric layer **135P** and IO layer **128P** along line C-C of FIG. 1D, when first HK gate dielectric layer **135P** includes HfO₂, and Ga and Zn dopants, and IO layer **128P** includes SiO₂, according to some embodiments. As shown in FIG. 1F, the peak concentration of Zn dopants (profile **139**) is higher than the peak concentration of Ga dopants (profile **140**).

FIG. 2 is a flow diagram of an example method **200** for fabricating PFET-NFET **102P-102N** with cross-sections as shown in FIGS. 1B-1C, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. 2 will be described with reference to the example fabrication process for fabricating PFET-NFET **102P-102N** as illustrated in FIGS. 3A-15B. FIGS. 3A-15B are cross-sectional views of PFET-NFET **102P-102N** along lines A-A and B-B of FIG. 1A at various stages of fabrication, according to various embodiments. Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **200** may not produce complete PFET-NFET **102P-102N**. Accordingly, it is understood that additional processes can be provided before, during, and after method **200**, and that some other processes may only be briefly described herein. Elements in FIGS. 3A-15B with the same annotations as elements in FIGS. 1A-1C are described above.

In operation **205**, polysilicon structures and S/D regions are formed on fin structures a PFET and NFET. For example, as shown in FIGS. 3A-3B polysilicon structures **312P-312N** and S/D regions **110P-110N** are formed on respective fin structure **106P-106N**, which are formed on substrate **104**. During subsequent processing, polysilicon structures **312P-312N** can be replaced in a gate replacement process to form gate structures **112P-112N**. After the formation of S/D regions **110P-110N**, ESLs **117A** and ILD layers **118** can be formed to form the structures of FIGS. 3A-3B.

Referring to FIG. 2, in operation **210**, gate openings are formed on the fin structures. For example, as shown in FIGS. 4A-4B, gate openings **412P-412N** are formed on respective fin structures **106P-106N**. The formation of gate openings **412P-412N** can include etching polysilicon structures **312P-312N** from the structures of FIGS. 3A-3B.

Referring to FIG. 2, in operations **215-235**, gate oxide structures are formed within the gate openings. For example, as described with reference to FIGS. 4A-12B, gate oxide structures **122P-122N** (shown in FIGS. 1B-1C) are formed within respective gate openings **412P-412N**.

Referring to FIG. 2, in operation **215**, IO layers are formed within the gate openings. For example, as shown in FIGS. 5A-5B, IO layers **128P-128N** are formed within gate openings **412P-412N**. In some embodiments, IO layers **128P-128N** can be formed by exposing the surfaces of fin structures **106P-106N** within respective gate openings **412P-412N** to an oxidizing ambient. The oxidizing ambient can include a combination of ozone (O₃), a mixture of ammonia hydroxide, hydrogen peroxide, and water, and/or a mixture of hydrochloric acid, hydrogen peroxide, water.

The subsequent formation of layers on IO layers **128P-128N** in operations **220-240** are described with reference to FIGS. 6A-13B, which are enlarged views of regions **103P-103N** shown in respective FIGS. 5A-5B.

Referring to FIG. 2, in operation **220**, first and second metal oxide layers are selectively formed on the IO layer of the PFET. For example, as described with reference to FIGS.

6A-7B, first and second metal oxide layers 132P-133P are formed on IO layer 128P. The selective formation of first and second metal oxide layers 132P-133P can include sequential operations of (i) depositing a first metal oxide layer 132 on the structures of FIGS. 5A-5B to form the structures of FIGS. 6A-6B, (ii) depositing a second metal oxide layer 133 on first metal oxide layer 132, as shown in FIGS. 6A-6B, and (iii) selectively removing the portions of first and second metal oxide layers 132-133 on IO layer 128N by using a lithographic patterning process on the structures of FIGS. 6A-6B to form the structures of FIGS. 7A-7B. Dipole layer 130P induced between first metal oxide layer 132 and IO layer 128N is removed when the portions of first and second metal oxide layers 132-133 on IO layer 128N are removed. The deposition of first and second metal oxide layers 132-133 can include depositing about 0.5 nm to about 3 nm layers of oxides of metals (e.g., Ga, Al, In, Zn, Nb, Mo, W, or Ta) that have electronegativity values greater than the electronegativity values of metals or semiconductors (e.g., Hf, Zr, or Ti) in first HK gate dielectric layer 134P. In addition, the layers of oxides (e.g., Ga_2O_3 , Al_2O_3 , In_2O_3 , ZnO , NbO_2 , MoO_2 , WO_3 , or Ta_2O_5) can have an oxygen areal density greater than the oxygen areal density of the oxide material (e.g., HfO_2 , ZrO_2 , or TiO_2) in first HK gate dielectric layer 134P. In some embodiments, the deposition of first oxide metal layer 132 includes depositing a layer of transition metal oxide (e.g., ZnO , NbO_2 , MoO_2 , WO_3 , or Ta_2O_5) and the deposition of second metal oxide layer 133 includes depositing a layer of oxide (e.g., Ga_2O_3 , Al_2O_3 , In_2O_3) of a metal from group 13 of the periodic table.

Referring to FIG. 2, in operation 225, a first HK gate dielectric layer with a first layer portion on the second metal oxide layer and a second layer portion on the IO layer of the NFET is formed. For example, as shown in FIGS. 8A-8B, a first HK gate dielectric layer with a first layer portion 134P (also referred to as “first HK gate dielectric layer 134P”) on second metal oxide layer 133P and a second layer portion 134N* (also referred to as “first HK gate dielectric layer 134N”) on IO layer 128N is formed. In some embodiments, first HK gate dielectric layers 134P-134N* can be formed by depositing about 1 nm to about 2 nm of HfO_2 with an atomic layer deposition (ALD) process using hafnium chloride (HfCl_4) as a precursor at a temperature ranging from about 250° C. to about 350° C.

Referring to FIG. 2, in operation 230, a doping process is selectively performed on the second layer portion of the first HK gate dielectric layer. For example, as described with reference to FIGS. 9A-11B, a doping process is selectively performed on first HK gate dielectric layer 134N*. The selective doping process can include sequential operations of (i) depositing a dopant source layer 946 on the structures of FIGS. 8A-8B to form the structures of FIGS. 9A-9B, (ii) selectively removing the portion of dopant source layer 946 on first HK gate dielectric layer 134P by using a lithographic patterning process on the structures of FIGS. 9A-9B to form the structures of FIGS. 10A-10B, (iii) performing a drive-in anneal process on the structures of FIGS. 10A-10B to form doped first HK gate dielectric layer 134N and dipole layer 130N, as shown in FIG. 11B, and (iv) removing dopant source layer 946 from the structure of FIG. 10B to form the structures of FIGS. 11A-11B.

The drive-in anneal process can implant metal dopants into first HK gate dielectric layer 134N* through diffusion of metal atoms from dopant source layer 946 into first HK gate dielectric layer 134N*. The implanted metal dopants can induce the formation of dipole layer 130N. The drive-in anneal process can include annealing the structures of FIGS.

10A-10B at a temperature from about 600° C. to about 800° C. and at a pressure from about 1 torr to about 50 torr for a time period ranging from about 0.1 second to about 30 seconds. In some embodiments, the drive-in anneal process can include two anneal processes: (i) a soak anneal process at a temperature from about 600° C. to about 800° C. and at a pressure from about 1 torr to about 50 torr for a time period ranging from about 2 sec to about 60 sec, and (ii) a spike anneal process at a temperature from about 700° C. to about 800° C. for a time period ranging from about 0.1 second to about 2 seconds.

The deposition of dopant source layer 946 can include depositing a layer of oxide of a rare-earth metal (e.g., La, Y, Sc, Ce, Yb, Er, Dy, or Lu) that has an electronegativity value lower than the electronegativity values of metallic or semiconductor materials (e.g., Hf, Zr, or Ti) included in first HK gate dielectric layer 134N. In addition, the layer of oxide (e.g., lanthanum oxide (La_2O_3), yttrium oxide (Y_2O_3), scandium oxide (Sc_2O_3), cerium oxide (CeO_2), ytterbium oxide (Yb_2O_3), erbium oxide (Er_2O_3), dysprosium oxide (Dy_2O_3), or lutetium oxide (Lu_2O_3)) can have an oxygen areal density smaller than the oxygen areal density of the oxide material (e.g., HfO_2 , ZrO_2 , or TiO_2) included in first HK gate dielectric layer 134N.

Referring to FIG. 2, in operation 235, a second HK gate dielectric layer with first and second layer portions on the first and second layer portions of the first HK gate dielectric layer is formed. For example, as shown in FIGS. 12A-12B, a second HK gate dielectric layer with a first layer portion 136P (also referred to as “second HK gate dielectric layer 136P”) on first HK gate dielectric layer 134P and a second layer portion 136N (also referred to as “second HK gate dielectric layer 136N”) on first HK gate dielectric layer 134N is formed. In some embodiments, second HK gate dielectric layers 136P-136N can be formed by depositing about 5 nm to about 8 nm of HfO_2 with an atomic layer deposition (ALD) process using hafnium chloride (HfCl_4) as a precursor at a temperature ranging from about 250° C. to about 350° C. In some embodiments, second HK gate dielectric layers 136P-136N can be deposited thicker than first HK gate dielectric layers 134P-134N.

Referring to FIG. 2, in operation 240, a pWFM layer is selectively formed on the first layer portion of the second HK gate dielectric layer and an nWFM layer is selectively formed on the second layer portion of the second HK gate dielectric layer. For example, as shown in FIGS. 13A-13B, pWFM layer 124P is selectively formed on second HK gate dielectric layer 136P and nWFM layer 124N is selectively formed on second HK gate dielectric layer 136N. The selective formation of pWFM 124P and nWFM layers 124N can be performed on the structures of FIGS. 12A-12B using lithographic patterning processes to form the structures of FIGS. 13A-13B.

Referring to FIG. 2, in operation 245, gate metal fill layers are formed on the pWFM layer and the nWFM layer. For example, as described with reference to FIGS. 14A-15B, gate metal fill layers 126P and 126N are formed on respective pWFM layer 124P and nWFM layer 124N. The formation of gate metal fill layers 126P-126N can include sequential operations of (i) depositing a conductive layer 1426 on the structures of FIGS. 13A-13B to fill gate openings 412P-412N and form the structures of FIGS. 14A-14B, and (ii) performing a chemical mechanical polishing (CMP) process on the structures of FIGS. 14A-14B to form the structures of FIGS. 15A-15B with top surfaces of gate structures 112P-112N substantially coplanar with a top surface of ILD layer 118. In some embodiments, after the CMP

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process, gate capping layers (not shown) on gate structures **112P-112N** and contact structures on gate structures **112P-112N** and S/D regions **110P-110N** can be formed.

FIG. **16** is a flow diagram of an example method **1600** for fabricating PFET-NFET **102P-102N** with cross-sections as shown in FIGS. **1D-1E**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. **16** will be described with reference to the example fabrication process for fabricating PFET-NFET **102P-102N** as illustrated in FIGS. **17A-26B**. FIGS. **17A-26B** are cross-sectional views of PFET-NFET **102P-102N** along lines A-A and B-B of FIG. **1A** at various stages of fabrication, according to various embodiments. Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **1600** may not produce complete PFET-NFET **102P-102N**. Accordingly, it is understood that additional processes can be provided before, during, and after method **1600**, and that some other processes may only be briefly described herein. Elements in FIGS. **17A-26B** with the same annotations as elements in FIGS. **1A-1E** are described above.

Referring to FIG. **16**, operations **1605-1615** are similar to operations **205-215** of FIG. **2**. After operation **1615**, structures similar to the structures of FIGS. **5A-5B** are formed. The subsequent formation of layers on IO layers **128P-128N** (shown in FIGS. **5A-5B**) in operations **1620-1640** are described with reference to FIGS. **17A-24B**, which are enlarged views of regions **103P-103N** shown in respective FIGS. **5A-5B**.

Referring to FIG. **16**, in operations **1615-1635**, gate oxide structures are formed within the gate openings. For example, as described with reference to FIGS. **17A-23B**, gate oxide structures **122P-122N** (shown in FIGS. **1D-1E**) are formed within respective gate openings **412P-412N** (shown in FIGS. **4A-4B**).

Referring to FIG. **16**, in operation **1620**, a first HK gate dielectric layer with first and second layer portions on respective IO layers of the PFET and NFET is formed. For example, as shown in FIGS. **17A-17B**, a first HK gate dielectric layer with a first layer portion **134P*** on IO layer **128P** and a second layer portion **134N*** on IO layer **128N** is formed. In some embodiments, first and second layer portions **134P*-134N*** can be formed by depositing about 1 nm to about 2 nm of HfO₂ with an atomic layer deposition (ALD) process using hafnium chloride (HfCl₄) as a precursor at a temperature ranging from about 250° C. to about 350° C.

Referring to FIG. **16**, in operation **1625**, a doping process is selectively performed on the first layer portion of the first HK gate dielectric layer. For example, as described with reference to FIGS. **18A-20B**, a doping process is selectively performed on first layer portion **134P***. The selective doping process can include sequential operations of (i) depositing a first dopant source layer **1848** on the structures of FIGS. **17A-17B**, as shown in FIGS. **18A-18B**, (ii) depositing a second dopant source layer **1850** on first dopant source layer **1848**, as shown in FIGS. **18A-18B**, (iii) selectively removing the portions of first and second dopant source layers **1848-1850** on second layer portion **134N*** by using a lithographic patterning process on the structures of FIGS. **18A-18B** to form the structures of FIGS. **19A-19B**, (iv) performing a drive-in anneal process on the structures of FIGS. **19A-19B** to form doped first HK gate dielectric layer **135P** and dipole layer **130P**, as shown in FIG. **20A**, and (iv) removing first and second dopant source layers **1848-1850** from the structure of FIG. **19** to form the structures of FIGS. **20A-20B**.

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The drive-in anneal process can implant metal dopants into first layer portion **134P*** through diffusion of metal atoms from first and second dopant source layers **1848-1850** into first layer portion **134P***. The implanted metal dopants can induce the formation of dipole layer **130P**. The drive-in anneal process can include annealing the structures of FIGS. **19A-19B** at a temperature from about 600° C. to about 800° C. and at a pressure from about 1 torr to about 50 torr for a time period ranging from about 0.1 second to about 30 seconds. In some embodiments, the drive-in anneal process can include two anneal processes: (i) a soak anneal process at a temperature from about 600° C. to about 800° C. for a time period ranging from about 2 sec to about 60 sec and (ii) a spike anneal process at a temperature from about 700° C. to about 800° C. for a time period ranging from about 0.1 second to about 2 seconds.

FIG. **27** shows the Hf, Zn, Ga, O₂, and Si concentration profiles **138**, **139**, **140**, **142**, and **144** along line D-D of FIG. **19A** before the drive-in anneal process, when first layer portion **134P*** includes HfO₂, first dopant source layer **1848** includes ZnO, second dopant source layer **1850** includes Ga₂O₃, and IO layer **128P** includes SiO₂, according to some embodiments. FIG. **28** shows the Hf, Zn, Ga, O₂, and Si concentration profiles **138**, **139**, **140**, **142**, and **144** along line D-D of FIG. **19A** after the drive-in anneal process. Prior to the drive-in anneal process, the peak concentrations of Zn and Ga atoms are within the respective regions of first and second dopant source layers **1848** and **1850**, as shown in FIG. **27**. Following the drive-in anneal process, the peak concentrations of Zn and Ga atoms shifts to the region of first HK gate dielectric layer **135P**, as shown in FIG. **28**, thus indicating the implantation of Zn and Ga dopants in first HK gate dielectric layer **135P**. The peak concentration of Zn atoms in first HK gate dielectric layer **135P** is higher and closer to interface **135P-128P** than the peak concentration of Ga atoms in first HK gate dielectric layer **135P**. Such difference between the peak concentrations of Zn and Ga atoms can be achieved by (i) using first dopant source layer **1848** with higher concentration of Zn atoms than the concentration of Ga atoms in second dopant source layer **1850**, and (ii) depositing first dopant source layer **1848** closer to first layer **134P*** than second dopant source layer **1850** to achieve threshold voltages below about 50 mV (e.g., about 30 mV or 20 mV).

The deposition of first and second dopant source layers **1848-1850** can include depositing layers of oxides of metals (e.g., Zn, Nb, Mo, W, Ta, Ga, Al, or In) that have electronegativity values greater than the electronegativity values of metals or semiconductors (e.g., Hf, Zr, or Ti) in first HK gate dielectric layer **135P**. In addition, the layers of oxides (e.g., ZnO, NbO₂, MoO₂, WO₃, or Ta₂O₅, Ga₂O₃, Al₂O₃, or In₂O₃) can have oxygen areal densities greater than the oxygen areal density of the oxide material (e.g., HfO₂, ZrO₂, or TiO₂) in first HK gate dielectric layer **135P**. The deposition of first and second dopant source layers **1848-1850** can include depositing layers of oxides different from each other. In some embodiments, second dopant source layer **1850** may not be deposited to dope first HK gate dielectric layer with the same metal dopants. In some embodiments, a nitride capping layer (not shown) can be deposited on second dopant source layer **1850** or on first dopant source layer **1848** in the absence of second dopant source layer **1850** to protect the underlying layers from thermal degradation during the drive-in anneal process.

Referring to FIG. **16**, in operation **1630**, a doping process is selectively performed on the second layer portion of the first HK gate dielectric layer. For example, as described with

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reference to FIGS. 21A-23B, a doping process is selectively performed on second layer portion 134N*. The selective doping process can include sequential operations of (i) depositing a dopant source layer 946 on the structures of FIGS. 20A-20B to form the structures of FIGS. 21A-21B, (ii) selectively removing the portion of dopant source layer 946 on first HK gate dielectric layer 135P by using a lithographic patterning process on the structures of FIGS. 21A-21B to form the structures of FIGS. 22A-22B, (iii) performing a drive-in anneal process on the structures of FIGS. 22A-22B to form doped first HK gate dielectric layer 134N and dipole layer 130N, as shown in FIG. 23B, and (iv) removing dopant source layer 946 from the structure of FIG. 22B. The deposition of dopant source layer 946 and the drive-in anneal process is similar to those described in operation 230.

Referring to FIG. 16, in operation 1635, a second HK gate dielectric layer with first and second layer portions on the doped first and second layer portions of the first HK gate dielectric layer is formed. For example, as shown in FIGS. 23A-23B, a second HK gate dielectric layer with a first layer portion 136P (also referred to as "second HK gate dielectric layer 136P") on first HK gate dielectric layer 135P and a second layer portion 136N (also referred to as "second HK gate dielectric layer 136N") on first HK gate dielectric layer 134N is formed. Second HK gate dielectric layers 136P-136N can be formed in an operation similar to that described in operation 235.

Referring to FIG. 16, in operation 1640, a pWFM layer is selectively formed on the first layer portion of the second HK gate dielectric layer and an nWFM layer is selectively formed on the second layer portion of the second HK gate dielectric layer. For example, as shown in FIGS. 24A-24B, pWFM layer 124P is selectively formed on second HK gate dielectric layer 136P and nWFM layer 124N is selectively formed on second HK gate dielectric layer 136N. The selective formation of pWFM 124P and nWFM layers 124N can be performed on the structures of FIGS. 23A-23B using lithographic patterning processes to form the structures of FIGS. 24A-24B.

Referring to FIG. 16, in operation 1645, gate metal fill layers are formed on the pWFM layer and the nWFM layer. For example, as described with reference to FIGS. 25A-26B, gate metal fill layers 126P and 126N are formed on respective pWFM layer 124P and nWFM layer 124N. The formation of gate metal fill layers 126P-126N is similar to that described in operation 245.

The present disclosure provides example multi-V_t devices with FETs (e.g., PFET 102P and NFET 102N) having ultra-low threshold voltages (e.g., about 20 mV to about 100 mV) different from each other and provides example methods of forming such FETs on the same substrate. The example methods form NFETs and PFETs with WFM layers (e.g., pWFM layer 124P and nWFM layer 124N) of similar thicknesses, but with ultra-low and/or different threshold voltages, on the same substrate. These example methods can be more cost-effective (e.g., cost reduced by about 20% to about 30%) and time-efficient (e.g., time reduced by about 15% to about 20%) in manufacturing reliable FET gate structures with lower and/or different threshold voltages than other methods of forming FETs with similar dimensions and threshold voltages on the same substrate. In addition, these example methods can form FET gate structures with much smaller dimensions (e.g., thinner gate stacks) than other methods of forming FETs with similar threshold voltages.

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In some embodiments, NFETs and PFETs with different gate structure configurations (e.g., gate structures 112P-112N), but with similar WFM layer thicknesses, can be selectively formed on the same substrate to achieve ultra-low and/or different threshold voltages. The different gate structures can have HK gate dielectric layers (e.g., HK gate dielectric layers 135P and 134N) doped with different metallic dopants. The different metal dopants can induce dipoles of different polarities and/or concentrations in dipole layers (e.g., dipole layers 130P-130N) at interfaces between the HK gate dielectric layers and IO layers. The dipoles of different polarities and/or concentrations result in gate structures with different EWF values and threshold voltages. In some embodiments, HK gate dielectric layers can be doped with dopants of two different materials to induce dipoles of different materials that provide stronger electric fields and lower threshold voltages. Thus, controlling the dopant materials and/or concentrations in the HK gate dielectric layers can tune the EWF values of the NFET and PFET gate structures, and as a result can adjust the threshold voltages of the NFETs and PFETs without varying the WFM layer thicknesses. In some embodiments, instead of the doped HK gate dielectric layer, PFET gate structure can include dual metal oxide layers (e.g., first and second metal oxide layers 132P-133N) interposed between the HK gate dielectric and the IO layer to induce dipoles of different materials between the HK gate dielectric layer and the IO layer.

In some embodiments, a method includes forming a fin structure on a substrate, forming a gate opening on the fin structure, forming an interfacial oxide layer on the fin structure, forming a first dielectric layer over the interfacial oxide layer, forming a dipole layer between the interfacial oxide layer and the first dielectric layer, forming a second dielectric layer on the first dielectric layer, forming a work function metal (WFM) layer on the second dielectric layer, and forming a gate metal fill layer on the WFM layer. The dipole layer includes ions of first and second metals that are different from each other. The first and second metals have electronegativity values greater than an electronegativity value of a metal or a semiconductor of the first dielectric layer.

In some embodiments, a method includes forming first and second fin structures on a substrate, forming first and second gate openings on the first and second fin structures, respectively, forming a first dielectric layer with first and second layer portions formed within the first and second gate openings, respectively, selectively doping the first layer portion with first and second dopants that are different from each other, selectively doping the second layer portion with third dopants different from the first and second dopants, forming a second dielectric layer with first and second layer portions on the first and second layer portions of the first dielectric layer, and forming first and second gate metal fill layers over the first and second layer portions of the second dielectric layer, respectively. The first and second dopants have electronegativity values greater than an electronegativity value of a metal or a semiconductor of the first dielectric layer. The second dopants have an electronegativity value less than an electronegativity value of the metal or the semiconductor of the first dielectric layer.

In some embodiments, a semiconductor device includes a substrate, a fin structure disposed on the substrate, a semiconductor oxide layer disposed on the fin structure, a first metal oxide layer disposed on the semiconductor oxide layer, a second metal oxide layer disposed on the first metal oxide layer, a first dielectric layer disposed on the second metal oxide layer, a second dielectric layer disposed on the

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first dielectric layer, a work function metal (WFM) layer disposed on the second dielectric layer, and a gate metal fill layer on the WFM layer. The second metal oxide layer is different from the first metal oxide layer. Metals of the first and second metal oxide layers have electronegativity values greater than an electronegativity value of a metal or a semiconductor the first dielectric layer.

The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device, comprising:
 - a substrate;
 - a fin structure disposed on the substrate;
 - a first metal oxide layer disposed on the fin structure;
 - a second metal oxide layer disposed on the first metal oxide layer, wherein the second metal oxide layer is different from the first metal oxide layer;
 - a first dielectric layer disposed on the second metal oxide layer, wherein metals of the first and second metal oxide layers have electronegativity values greater than an electronegativity value of a metal or a semiconductor of the first dielectric layer; and
 - a gate metal fill layer on the first dielectric layer.
2. The semiconductor device of claim 1, further comprising a second dielectric layer disposed on the first dielectric layer.
3. The semiconductor device of claim 1, further comprising first and second dipoles disposed between the fin structure and the first metal oxide layer, wherein the first and second dipoles are different from each other.
4. The semiconductor device of claim 1, further comprising:
 - a first concentration of first dipoles disposed between the fin structure and the first metal oxide layer; and
 - a second concentration of second dipoles disposed between the fin structure and the first metal oxide layer, wherein:
 - the first and second dipoles are different from each other; and
 - the first and second concentrations are different from each other.
5. The semiconductor device of claim 1, further comprising first and second dipoles disposed between the fin structure and the first metal oxide layer, wherein the first dipole comprises a metal atom of the first metal oxide layer and the second dipole comprises a metal atom of the second metal oxide layer.
6. The semiconductor device of claim 1, further comprising first and second dipoles disposed between the fin structure and the first metal oxide layer, wherein the first dipole comprises an ion of a transition metal and the second dipole comprises an ion of a metal from group 13 of the periodic table.
7. The semiconductor device of claim 1, wherein the first metal oxide layer comprises an oxide of a transition metal.

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8. The semiconductor device of claim 1, wherein the second metal oxide layer comprises an oxide of a metal from group 13 of the periodic table.

9. The semiconductor device of claim 1, wherein the first and second metal oxide layers have oxygen areal densities greater than an oxygen areal density of the first dielectric layer.

10. The semiconductor device of claim 1, wherein a thickness of the first metal oxide layer is greater than a thickness of the second metal oxide layer.

11. A semiconductor device, comprising:

a substrate;

a fin structure disposed on the substrate;

a first dielectric layer disposed on the fin structure, wherein the first dielectric layer comprises first and second metal dopants different from each other, and wherein a first peak of a concentration profile of the first metal dopant is closer to a bottom surface of the first dielectric layer than a second peak of a concentration profile of the second metal dopant;

a second dielectric layer disposed on the first dielectric layer; and

a gate metal fill layer on the second dielectric layer.

12. The semiconductor device of claim 11, wherein the first and second metal dopants have electronegativity values greater than an electronegativity value of a metal or a semiconductor of the first dielectric layer.

13. The semiconductor device of claim 11, wherein the first metal dopant comprises a transition metal, and wherein the second metal dopant comprises a metal from group 13 of the periodic table.

14. The semiconductor device of claim 11, wherein the first metal dopant comprises zinc, and wherein the second metal dopant comprises gallium.

15. The semiconductor device of claim 11, further comprising first and second dipoles disposed between the fin structure and the first dielectric layer, wherein the first and second dipoles are different from each other.

16. The semiconductor device of claim 11, further comprising first and second dipoles disposed between the fin structure and the first dielectric layer, wherein the first dipole comprises an ion of a transition metal and the second dipole comprises an ion of a metal from group 13 of the periodic table.

17. A method, comprising:

forming a fin structure on a substrate;

depositing a first dielectric layer on the fin structure;

forming a dipole layer between the first dielectric layer and the fin structure, wherein the dipole layer comprises ions of first and second metals that are different from each other;

depositing a second dielectric layer on the first dielectric layer; and

depositing a gate metal fill layer on the second dielectric layer.

18. The method of claim 17, wherein forming the dipole layer comprises doping the first dielectric layer with first and second metal dopants different from each other.

19. The method of claim 17, wherein forming the dipole layer comprises depositing, on the first dielectric layer, first and second dopant source layers different from each other.

20. The method of claim 17, further comprising depositing, on the first dielectric layer, first and second metal oxide layers different from each other.

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